Notes on Gravitation and Relativity

Nesar Ali Titumir

Florida State University

October 7, 2026

Contents

1 Some Mathematical Preliminaries **2**

1.1 Introduction . . . 2

1.2 Revisiting Calculus Notions . . . 2

2 Manifolds **10**

2.1 Topological Space . . . 10

2.2 Hausdorff Topological Space . . . 13

2.3 Inheriting a topology . . . 15

2.4 Top. Manifold . . . 16

2.5 Compatibility of charts . . . 16

2.6 Maps between manifolds . . . 17

2.7 Algebra of Forms . . . 19

2.7.1 Wedge Product . . . 20

2.7.2 Exterior Derivative . . . 22

2.8 Integration on a Manifold . . . 22

2.8.1 Orientation and Volume Form . . . 22

2.9 Non-coordinate Basis and Tetrad Formalism . . . 24

2.9.1 Connection 1-form and Ricci Rotation Coefficient . . . 27

2.9.2 Tensor Valued Forms and Exterior Derivatives . . . 29

3 Derivatives on Manifolds **31**

3.1 Introduction . . . 31

3.2 Lie Derivative . . . 31

3.3 Covariant Derivative . . . 32

4 Submanifolds and Hypersurfaces **33**

4.1 Introduction . . . 33

4.2 Submanifolds . . . 33

4.2.1 Imbedding . . . 33

4.2.2 Tensors Tangent and Normal to S . . . 34

4.2.3 Sticking to Only M-Tensors . . . 37

4.2.4 S-smoothness and M-smoothness . . . 38
4.2.5 h_{ab} . . . 38
4.2.6 Derivatives on S . . . 40
4.2.7 Extrinsic Curvature . . . 44
4.2.8 Gauss Codazzi Equations . . . 44
4.3 Hypersurfaces . . . 49
4.3.1 Gauss-Codazzi eqns for hypersurface . . . 51
4.3.2 Integration on Hypersurfaces . . . 52

5 Gravitational Waves **55**
5.1 The Harmonic Gauge . . . 55
5.2 Perturbative Approach . . . 56

Introduction

Introduction

Chapter 1

Some Mathematical Preliminaries

1.1 Introduction

This chapter introduces some fundamental mathematics that will be needed in the later chapters. The terms appearing in this chapter should be familiar to any student of physics. However, due to the intuitive way these terms are usually presented in any first course of calculus at the university freshman level, students are not always familiar with the more "mathematical" way of presenting these concepts. Since in the later study of basic differential geometry these will work as a necessary prerequisite vocabulary, it is better to be familiar with the language a bit.

1.2 Revisiting Calculus Notions

Let $U \subseteq \mathbb{R}^n$ and $f : U \longrightarrow \mathbb{R}$. We define the partial derivatives as follows:—

$$\left.\frac{\partial f}{\partial x^i}\right|_a := \lim_{h\to 0} \frac{f\left(a^1, \ldots, a^i + h, \ldots, a^n\right) - f\left(a^1, \ldots, a^i, \ldots, a^n\right)}{h}$$

If $\left.\frac{\partial f}{\partial x^i}\right|_a$ exists for all i $(1 \leq i \leq n)$ for each point $a \in U$, then we have n new functions in U. If these functions $\frac{\partial f}{\partial x^i}$ are also continuous on U, then we say that f is continuosly differentiable on U, i.e, $f \in C^1(U)$.

Existence of partial derivatives at point a is not enough to guarantee the differentiability of f at a. To illustrate this point, we use an example. Consider the function, defined on $\mathbb{R}^2$, by

$$\begin{cases} f(x,y) = \frac{xy}{x^2+y^2} \\ f(0,0) = 0 \end{cases}$$

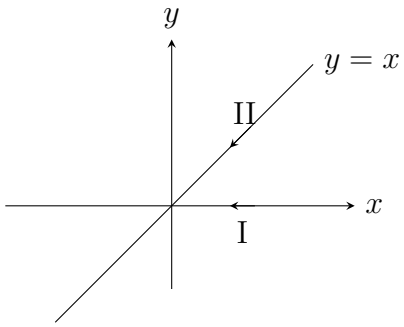
We evalute the following,

$$\begin{aligned} \left.\frac{\partial f}{\partial x}\right|_{(0,0)} &= \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} \\ &= \lim_{h\to 0} \frac{0-0}{h} = 0 \end{aligned}$$

Similarly, $\left.\frac{\partial f}{\partial y}\right|_{(0,0)} = 0$

So, both of the partial derivaties exist at the point $(0,0)$. But the function is still not differentiable. To prove that, it suffices to show that the function is not continuous at $(0,0)$.

To decide for continuity, we first try to find out the limit of $f(x,y)$ at $(0,0)$. Considering the two different paths as in the picture



$$\begin{aligned} \lim_{\substack{(x,y)\to(0,0) \\ \text{along path II}}} f(x,y) &= \lim_{x\to 0} \frac{x \cdot x}{x^2 + x^2} \\ &= \lim_{x\to 0} \frac{x^2}{2x^2} \\ &= \frac{1}{2} \end{aligned}$$

$$\lim_{\substack{(x,y)\to(0,0)\\ \text{along I}}} f(x,y) = 0$$

$\therefore$ limit along I $\neq$ limit along II

$\therefore$ The limit DNE at the point $(0,0)$ and so $f(x,y)$ is not continuos at $(0,0)$. Hence it is not differentiable at the origin even though both of the partial derivatives exist.

Let, $f : U \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}$

If f has continuous first partial derivatives throughout U, then $f \in C^1(U)$.

Definition 1.1. We consider $f \in C^r(U)$ iff for all $j \leq r$,

$$\left.\frac{\partial^j f}{\partial x^{i_1}\partial x^{i_2}\ldots\partial x^{i_j}}\right|_a \qquad (1 \leq i_1,\ldots,i_j \leq n) \text{ exists}$$

for all $a \in U$ and they are continuos on U.

Here the order of the derivatives doesnt matter.

A function $f : U \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}$ is said to be differentiable at a point a, iff there exists a linear expression $\sum_{i=1}^n b_i\,(x^i - a^i)$ which satisfies the following

$$\lim_{x\to a} \frac{f(x) - f(a) - \sum_{i=1}^n b_i\,(x^i - a^i)}{\|x-a\|} = 0$$

So, we can say that $f(a) + \sum_{i=1}^n b_i\,(x^i - a^i)$ approximates $f(x)$ near the point a.

Using the above expression, or, you can say keeping it at the back of our mind, we can come to another def^{n} of differentiability. Here, a function is said to be differentiable at a, if there exists constants $b_1,\ldots,b_n$ and a function $r(x,a)$ such that

$$f(x) = f(a) + \sum_{i=1}^{n} b_i\left(x^i - a^i\right) + \|x-a\|r(x,a)$$
$$\text{and } \lim_{x\to a} r(x,a) = 0$$

When a function $f : U \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}$ is differentiable at a, then f is continuous at a and $b_i = \frac{\partial f}{\partial x^i}\Big|_a$

$$\text{So, } f(x) = f(a) + \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}\Big|_a \left(x^i - a^i\right) + r(x,a)\|x-a\|$$

Keep in mind that, f is differentiable at a, merely guarantees the existence of $\frac{\partial f}{\partial x^i}\Big|_a$. It doesnt guarantee that f is C^1 at the point a.

If f is C^1 in a neighborhood of a, then f is guaranteed to be differentiable at that neighbourhood.

Thus if $f \in C^1(U)$ where U is an open set, then f is differentiable on U.

Let's generalize the notions learned so far.

Assume that, $f : U \subseteq \mathbb{R}^n \longrightarrow V \subseteq \mathbb{R}^m$

The first thing that we must make clear is as follows:—

1. $f(x) = (f^1(x), f^2(x), \ldots, f^m(x))$

   we call the individual functions f^i to be coordinate functions. We define them as follows. Suppose $\pi^i : \mathbb{R}^m \longrightarrow \mathbb{R}$ is a function given by $\pi^i\left(x^1, x^2, \ldots, x^m\right) = x^i$.

   Now, $f^i := \pi^i \circ f : U \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}$

   So it's clear from the above that,

$$f(x) = \left(f^1(x), \ldots, f^m(x)\right)$$

2. f is continuous on U if $f^1, \ldots, f^m$ are continuous on U.

   f is differentiable on U if $f^1, \ldots, f^m$ are

   f is $C^r(U)$ if $f^1, \ldots, f^m$ are $C^r(U)$

For ease of computation, we assume that the points of $\mathbb{R}^n$ or $\mathbb{R}^m$ are given by column matrices. So, for instance,

$$a \in \mathbb{R}^n \text{ and } a = \begin{bmatrix} a^1 \\ a^2 \\ \vdots \\ a^n \end{bmatrix}$$

In this case

$$f(x) = \begin{bmatrix} f^1(x) \\ f^2(x) \\ f^3(x) \\ \vdots \\ f^m(x) \end{bmatrix}$$

We give another def^{n} for differentiability.

$f : U \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^n$ is differentiable at a point $a \in U$ if there exists a matrix A and a function $r(x, a)$ that satisfies the following two conditions:—

$$f(x) = f(a) + A\,(x - a) + r(x, a)\|x - a\|$$
$$\text{and } \lim_{x \to a} r(x, a) = 0$$

If f is differentiable at the point a, then A is the unique Jacobian matrix at a.

Now comes the def^{n} of the Jacobian matrix.

For any point $p \in U$, we define the following:—

$$\text{Jac}f(p) = \left.\frac{\partial\,(f^1, \ldots, f^m)}{\partial\,(x^1, \ldots, x^n)}\right|_p := \begin{bmatrix} \left.\frac{\partial f^1}{\partial x^1}\right|_p & \left.\frac{\partial f^1}{\partial x^2}\right|_p & \cdots & \left.\frac{\partial f^1}{\partial x^n}\right|_p \\ \left.\frac{\partial f^2}{\partial x^1}\right|_p & \left.\frac{\partial f^2}{\partial x^2}\right|_p & \cdots & \left.\frac{\partial f^2}{\partial x^n}\right|_p \\ \vdots & & \ddots & \vdots \\ \left.\frac{\partial f^m}{\partial x^1}\right|_p & \cdots & \cdots & \left.\frac{\partial f^m}{\partial x^n}\right|_p \end{bmatrix}$$

This is the Jacobian matrix of the function f at the point P. This matrix defines a map $\left.Df\right|_p : U \longrightarrow \mathbb{R}^m$. At each point of U, we get different different maps.

So, for the function f (when it is differentiable at the point a),

$$f(x) = f(a) + \left(\left.Df\right|_a\right)(x - a) + r(x, a)\|x - a\|$$

$Df\Big|_a$ acting on $(x-a)$ which basially is

$$\text{Jac}f(a)\begin{bmatrix} x^1-a^1 \\ x^2-a^2 \\ \vdots \\ x^n-a^n \end{bmatrix}$$

Now, we come to prove chain rule in its most general version.

$$\begin{aligned} f &: U \subseteq \mathbb{R}^n \longrightarrow V \subseteq \mathbb{R}^m \\ g &: V \subseteq \mathbb{R}^m \longrightarrow \mathbb{R}^k \\ \therefore g \circ f &: U \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^k \end{aligned}$$

The chain rule says,

$$Dg \circ f\Big|_p = Dg\Big|_{f(p)} \circ Df\Big|_p$$

Using the general case of the chain rule, we can derive important special cases:—

1.

$$\begin{aligned} f &: U \subseteq \mathbb{R}^n \longrightarrow V \subseteq \mathbb{R}^m \\ g &: V \subseteq \mathbb{R}^m \longrightarrow \mathbb{R}^k \\ g \circ f &: U \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^k \end{aligned}$$

$$\frac{\partial \left(g \circ f\right)^i}{\partial x^j}\bigg|_p = \frac{\partial g^i}{\partial u^k}\bigg|_{f(p)} \frac{\partial f^k}{\partial x^j}\bigg|_p$$

2.

$$f : U \subseteq \mathbb{R}^n \longrightarrow V \subseteq \mathbb{R}^m$$
$$g : V \subseteq \mathbb{R}^m \longrightarrow \mathbb{R}$$
$$\therefore g \circ f : U \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}$$

$$\left.\frac{\partial (g \circ f)}{\partial x^i}\right|_p = \left.\frac{\partial g}{\partial u^j}\right|_{f(p)} \left.\frac{\partial f^j}{\partial x^i}\right|_p$$

3.

$$f : U \subseteq \mathbb{R}^n \longrightarrow V \subseteq \mathbb{R}$$
$$g : V \subseteq \mathbb{R} \longrightarrow \mathbb{R}$$
$$\therefore g \circ f : U \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}$$

$$\left.\frac{\partial (g \circ f)}{\partial x^i}\right|_p = \left.\frac{\partial g}{\partial u}\right|_{f(p)} \left.\frac{\partial f}{\partial x^i}\right|_p$$

4.

$$f : U \subseteq \mathbb{R} \longrightarrow V \subseteq \mathbb{R}$$
$$g : V \subseteq \mathbb{R} \longrightarrow \mathbb{R}$$
$$g \circ f : U \subseteq \mathbb{R} \longrightarrow \mathbb{R}$$

$$\left.\frac{d (g \circ f)}{dx}\right|_p = \left.\frac{dg}{du}\right|_{f(p)} \left.\frac{df}{dx}\right|_p$$

5.

$$f : U \subseteq \mathbb{R} \longrightarrow V \subseteq \mathbb{R}^n$$
$$g : V \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^m$$
$$g \circ f : U \subseteq \mathbb{R} \longrightarrow \mathbb{R}^m$$

$$\left.\frac{\partial \left(g \circ f\right)^i}{\partial x}\right|_p = \left.\frac{\partial g^i}{\partial u^k}\right|_{f(p)} \left.\frac{\partial f^k}{\partial x}\right|_p$$

6.

$$f : U \subseteq \mathbb{R} \longrightarrow V \subseteq \mathbb{R}^m$$
$$g : V \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}$$
$$g \circ f : U \subseteq \mathbb{R} \longrightarrow \mathbb{R}$$

$$\therefore \left.\frac{\partial \left(g \circ f\right)}{\partial x}\right|_p = \left.\frac{\partial g}{\partial u^i}\right|_{f(p)} \left.\frac{\partial f^i}{\partial x}\right|_p$$

If f and g are of class C^r (or smooth) on U and V, then $h = g \circ f$ is of class C^r (smooth) on U.

Chapter 2

Manifolds

2.1 Topological Space

Definition 2.1. Let M be a set. We define a topology $\mathcal{O}$ on M to be a collection of subsets of M such that

(i) $\emptyset, M \in \mathcal{O}$

(ii) $\mathcal{O}$ is closed under arbitrary union and finite intersection. i.e.

$$U, V \in \mathcal{O} \implies U \cap V \in \mathcal{O}$$
$$U_\alpha \in \mathcal{O}\,(\alpha \in A) \implies \bigcup_{\alpha \in A} U_\alpha \in \mathcal{O}$$

Definition 2.2. A set M equipped with a topology $\mathcal{O}$, i.e. $(M, \mathcal{O})$ is called a topological space.

Examples of topological spaces are as follows. For any set M, the collection $\mathcal{O} = \{\emptyset, M\}$ is a topology. This is called the chaotic topology. The collection $\mathcal{O} = P(M)$ (Power set of M) is called the discrete topology. Another example is $(M, \mathcal{O})$ with $M = \{1, 2, 3\}$ and $\mathcal{O} = \{\{1\}, \{2\}, \{1, 2\}, \{1, 2, 3\}, \emptyset\}$

Consider $\mathbb{R}^n$ where we define the "open ball/soft ball" of radius r around a point $p \in \mathbb{R}^n$ as:

$$B_r(p) := \left\{ x \in \mathbb{R}^n \;\middle|\; \sum_{i=1}^{n} |x^i - p^i|^2 < r^2 \right\}$$

With the help of this we can define a topology $\mathcal{O}_{std}$ on $\mathbb{R}^n$.

Definition 2.3. A set $U \in \mathcal{O}_{std}$ iff $U \subseteq \mathbb{R}^n$ and

$$p \in U \implies \exists r \in \mathbb{R}^+ : B_r(p) \subseteq U$$

Proposition 2.4. $(\mathbb{R}^n, \mathcal{O}_{std})$ is a topological space.

Proof. $\emptyset \in \mathcal{O}_{std}$ since $p \in \emptyset \implies \exists r \in \mathbb{R}^+ : B_r(p) \subseteq \emptyset$ (vacuously)
$\mathbb{R}^n \in \mathcal{O}_{std}$ since $p \in \mathbb{R}^n \implies \exists r = 1 \in \mathbb{R}^+ : B_r(p) \subseteq \mathbb{R}^n$
If $U, V \in \mathcal{O}_{std}$ then assume $p \in U \cap V$

$$\begin{aligned} p \in U \cap V &\implies p \in U \text{ and } p \in V \\ &\implies \exists r_1, r_2 \in \mathbb{R}^+ : B_{r_1}(p) \subseteq U \text{ and } B_{r_2}(p) \subseteq V \end{aligned}$$

Now, $r := \min(r_1, r_2)$

$$\begin{aligned} &\therefore B_r(p) \subseteq U \text{ and } B_r(p) \subseteq V \\ &\implies B_r(p) \subseteq U \cap V \\ &\therefore U, V \in \mathcal{O}_{std} \implies (U \cap V) \in \mathcal{O}_{std} \end{aligned}$$

Assume $U_\alpha \in \mathcal{O}_{std}$ $(\alpha \in A)$
Let $p \in \bigcup_{\alpha \in A} U_\alpha$

$$\begin{aligned} &\implies p \in U_\beta \text{ for some } \beta \in A \\ &\implies \exists r \in \mathbb{R}^+ : B_r(p) \subseteq U_\beta \\ &\implies \exists r \in \mathbb{R}^+ : B_r(p) \subseteq \bigcup_{\alpha \in A} U_\alpha \\ &\therefore U_\alpha \in \mathcal{O}_{std}\,(\alpha \in A) \implies \left(\bigcup_{\alpha \in A} U_\alpha \right) \in \mathcal{O}_{std} \end{aligned}$$

□

So, in short, we can say that $(\mathbb{R}^n, \mathcal{O}_{std})$ is a topological space.

Basis

The basic idea is to have a collection of subsets from which we can take unions and construct any set $U \in \mathcal{O}$.

Definition 2.5. A collection $\mathcal{B}$ of open sets is a basis for the topology $\mathcal{O}$ if given any set $U \in \mathcal{O}$ and a point $p \in U$, there is an open set $B \in \mathcal{B}$ such that $p \in B \subseteq U$.

An element of the basis $\mathcal{B}$ is called a basic open set.

Theorem 2.6. A collection of open sets $\mathcal{B}$ is a basis of $\mathcal{O}$ iff any open set U is a union of sets of $\mathcal{B}$.

Proof. ($\implies$) We assume that $\mathcal{B}$ is a basis for $\mathcal{O}$. So, if $U \in \mathcal{O}$, then for $p \in U$, $\exists B_p \in \mathcal{B}$ such that $p \in B_p \subseteq U$.

$$\therefore U = \bigcup_{p \in U} B_p$$

($\impliedby$) Since any open set U is a union of sets of $\mathcal{B}$, for any point $p \in U$, we have a set $B \in \mathcal{B}$ such that $p \in B \subseteq U$. □

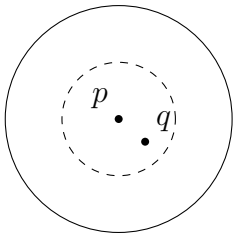
$\mathcal{B}_{rat}$ is the set of all open balls $B_r(p)$ where $r \in \mathbb{Q}^+$ and $p \in \mathbb{Q}^n$, i.e. rational centers and rational radius.

Theorem 2.7. The set $\mathcal{B}_{rat}$ is a basis for $\mathcal{O}_{std}$.

Proof. Take $U \in \mathcal{O}_{std}$

Now, for any point $p \in U$, there exists $r \in \mathbb{Q}^+$ such that $B_r(p) \subseteq U$.

Now, take a rational point $q \in B_{r/2}(p)$



So,

$$d(p, q) < \frac{r}{2}$$

Consider the ball $B_{r/2}(q)$

$$p \in B_{r/2}(q) \tag{2.1}$$

Let, $x \in B_{r/2}(q)$

$$\implies d(x,q) < \frac{r}{2}$$

Now,

$$\begin{aligned} &d(x,p) \leq d(x,q) + d(q,p) < \frac{r}{2} + \frac{r}{2} = r \\ &\implies d(x,p) < r \implies x \in B_r(p) \\ &\therefore p \in B_{r/2}(q) \subseteq B_r(p) \subseteq U \end{aligned}$$

□

So, we can say that $\mathcal{B}_{rat}$ is a basis for $\mathcal{O}_{std}$.

There is a curious thing to note here. The set $\mathcal{B}_{rat}$ of all the rational balls (balls with rational center and rational radii) is a countable set. Since each ball can be indexed by $\mathbb{Q}^n$ (a countable set) and $\mathbb{Q}^+$ (another countable set), we have a countable infinity of rational balls.

Definition 2.8. A topological space $(M, \mathcal{O})$ is called second countable iff $\mathcal{O}$ has a countable basis.

2.2 Hausdorff Topological Space

Definition 2.9. $(T, \mathcal{O}_T)$ is called a Hausdorff top. space :=

$$\begin{aligned} &\text{iff } \forall p, q \in T \text{ and } p \neq q \\ \implies &\exists U_p \in \mathcal{O}_T \text{ and } \exists U_q \in \mathcal{O}_T,\ p \in U_p \ \&\ q \in U_q \end{aligned}$$

$$\text{and } U_p \cap U_q = \emptyset$$

In short, for every pair of points, we have disjoint open sets that contain those points.

Definition 2.10. Let, $(M, \mathcal{O}_M)$ and $(N, \mathcal{O}_N)$ be top. sp. A map $f : M \longrightarrow N$ is called continuous with respect to $\mathcal{O}_M$ and $\mathcal{O}_N$ iff

$$V \in \mathcal{O}_N \implies \text{preim}_f(V) \in \mathcal{O}_M$$

i.e. preimage of open sets are open.

We have defined, for $f : M \longrightarrow N$ and $V \subseteq N$

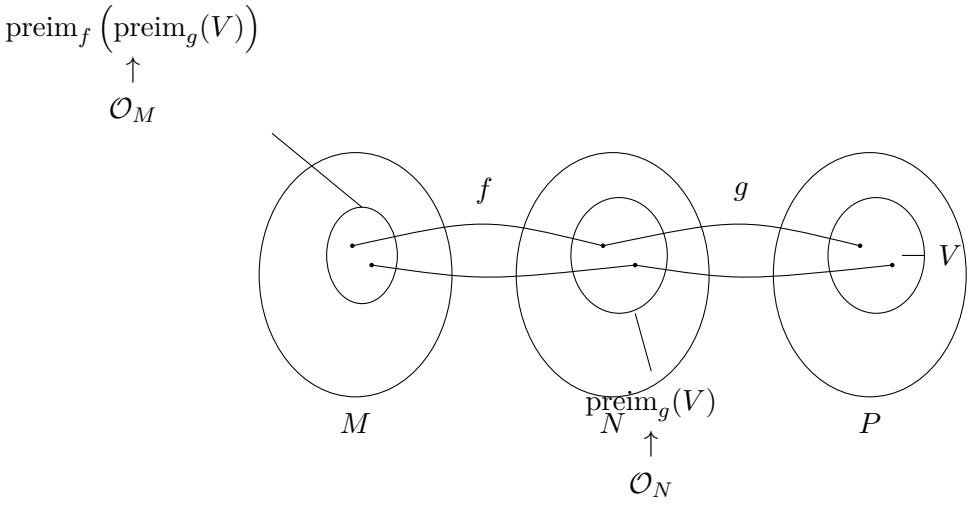
$$\text{preim}_f(V) = \{m \in M \mid f(m) \in V\}$$

Proposition 2.11. Let, $(M, \mathcal{O}_M)$, $(N, \mathcal{O}_N)$, $(P, \mathcal{O}_P)$ be three top. sp.
$f : M \longrightarrow N$, $g : N \longrightarrow P$ are continuous maps.
The map $h = g \circ f$ is also continuous w.r.t. $\mathcal{O}_M$ & $\mathcal{O}_P$.

Proof. Let, $V \in \mathcal{O}_P$
Now,

$$\begin{aligned}
\text{preim}_h(V) &= \{m \in M \mid h(m) \in V\} \\
&= \{m \in M \mid (g \circ f)(m) \in V\} \\
&= \{m \in M \mid f(m) \in \text{preim}_g(V)\} \\
&= \text{preim}_f\left(\text{preim}_g(V)\right) \\
&\in \mathcal{O}_M
\end{aligned}$$

□



Use the following for visualization.

$f : M \longrightarrow N$ is a homeomorphism between M and N if f and f^{-1} both are continuous w.r.t. $\mathcal{O}_M$ and $\mathcal{O}_N$.

2.3 Inheriting a topology

Let $(T, \mathcal{O})$ be a top. sp. and $S \subseteq T$.

$$\mathcal{O}|_S := \{U \cap S \mid U \in \mathcal{O}\}$$

$(S, \mathcal{O}|_S)$ is a top. sp.

Proof.

$$\emptyset = \emptyset \cap S \text{ and } \emptyset \in \mathcal{O} \implies \emptyset \in \mathcal{O}|_S$$
$$S = S \cap M \text{ and } M \in \mathcal{O} \implies S \in \mathcal{O}|_S$$

Let, S_1 and $S_2 \in \mathcal{O}|_S$

$$\implies \exists U_1, U_2 \in \mathcal{O} \text{ such that } S_1 = U_1 \cap S,\ S_2 = U_2 \cap S$$
$$\implies U_1 \cap U_2 \in \mathcal{O} \implies (U_1 \cap U_2) \cap S \in \mathcal{O}|_S$$
$$\implies (U_1 \cap S) \cap (U_2 \cap S) \in \mathcal{O}|_S \implies S_1 \cap S_2 \in \mathcal{O}|_S$$

Let, $\alpha \in I$ and $S_\alpha \in \mathcal{O}|_S \ \forall \alpha \in I$

$$\implies \forall \alpha \ \exists U_\alpha \in \mathcal{O} \text{ such that } S_\alpha = U_\alpha \cap S$$
$$\implies \bigcup_{\alpha \in I} U_\alpha \in \mathcal{O} \implies \left(\bigcup_{\alpha \in I} U_\alpha \right) \cap S \in \mathcal{O}|_S$$
$$\implies \bigcup_{\alpha \in I} (U_\alpha \cap S) \in \mathcal{O}|_S \implies \bigcup_{\alpha \in I} S_\alpha \in \mathcal{O}|_S$$

□

2.4 Top. Manifold

A Hausdorff, second countable top. space $(M, \mathcal{O})$ is called a d-dimensional topological manifold if for every point $p \in M$, there is an open set U such that $p \in U$ and there is a homeomorphism $x : U \longrightarrow x(U) \subseteq \mathbb{R}^d$

where $x(U) \in \mathcal{O}_{std}$

Note:- The map x is bi-continuous w.r.t. the top. spaces $(U, \mathcal{O}|_U)$ and $\left(x(U), \mathcal{O}_{std}|_{x(U)}\right)$

Definition 2.12. Chart :=

Let $(M, \mathcal{O})$ be a d-dim top. mfd. Then the pair (U, x) where $U \in \mathcal{O}$ and x is a homeomorphism $x : U \longrightarrow x(U) \subseteq \mathbb{R}^d$, is called a chart of the manifold.

Definition 2.13. Atlas :=

An atlas of a manifold M is a collection of charts

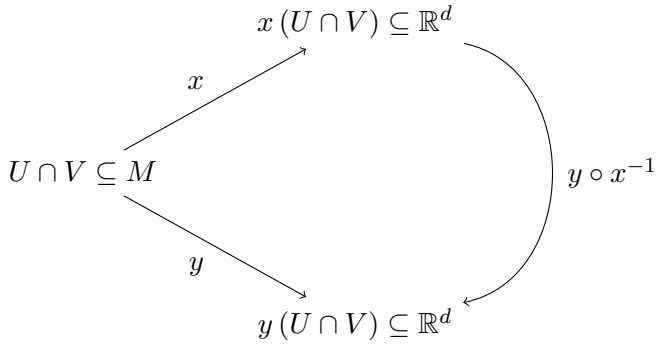
$$\mathcal{A} := \{(U_\alpha, x_\alpha) \mid \alpha \in A\} \text{ such that}$$
$$\bigcup_{\alpha \in A} U_\alpha = M$$

2.5 Compatibility of charts

Two charts (U, x) and (V, y) are $\mathcal{C}$-compatible is either $U \cap V \neq \emptyset$ are the maps

$$x \circ y^{-1} : y\,(U \cap V) \longrightarrow x\,(U \cap V)$$
$$y \circ x^{-1} : x\,(U \cap V) \longrightarrow y\,(U \cap V)$$

is undergraduate-$\mathcal{C}$



"$x \circ y^{-1}$ and $y \circ x^{-1}$ are called chart-transition maps"

Charts of a top. manifold is automatically $\mathcal{C}^0$-compatible

Definition 2.14. $\mathcal{C}$-Atlas :=

An atlas $\mathcal{A}$ of a top. manifold is called a $\mathcal{C}$-Atlas if the charts of $\mathcal{A}$ are $\mathcal{C}$-compatible.

Definition 2.15. $\mathcal{C}^k$-manifold:

A $\mathcal{C}^k$-manifold is a triple $(M, \mathcal{O}, \mathcal{A})$ where $(N, \mathcal{O})$ is a top. manifold and $\mathcal{A}$ is a maximal $\mathcal{C}^k$-atlas.

Definition 2.16. Smooth-manifold:

A $\mathcal{C}^\infty$-manifold is called a smooth manifold.

From now on we are going to explicitly consider smooth manifolds.

2.6 Maps between manifolds

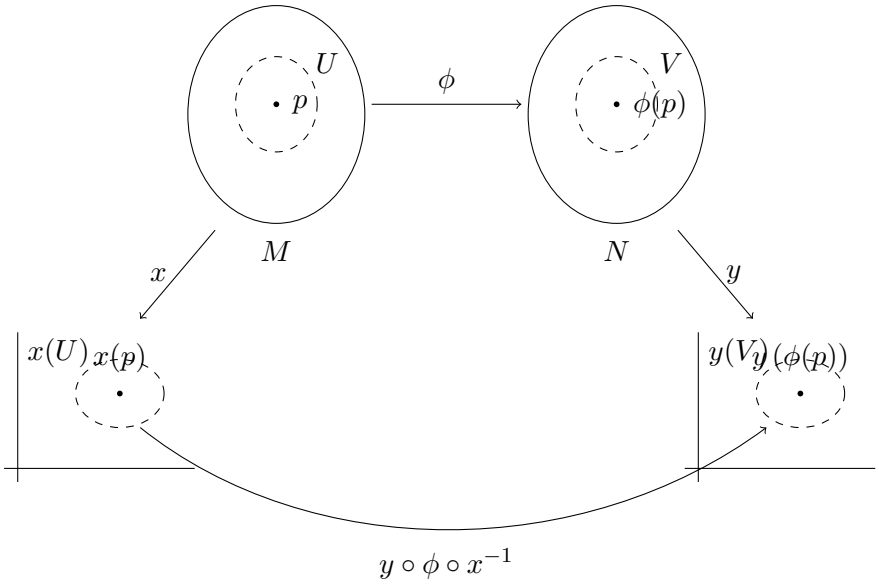
Let, $(M, \mathcal{O}, \mathcal{A}_M)$ and $(N, \mathcal{O}, \mathcal{A}_N)$ be manifolds.

$\phi : M \longrightarrow N$ is a map from M to N.

ϕ will be smooth map between M and N if the following holds:—

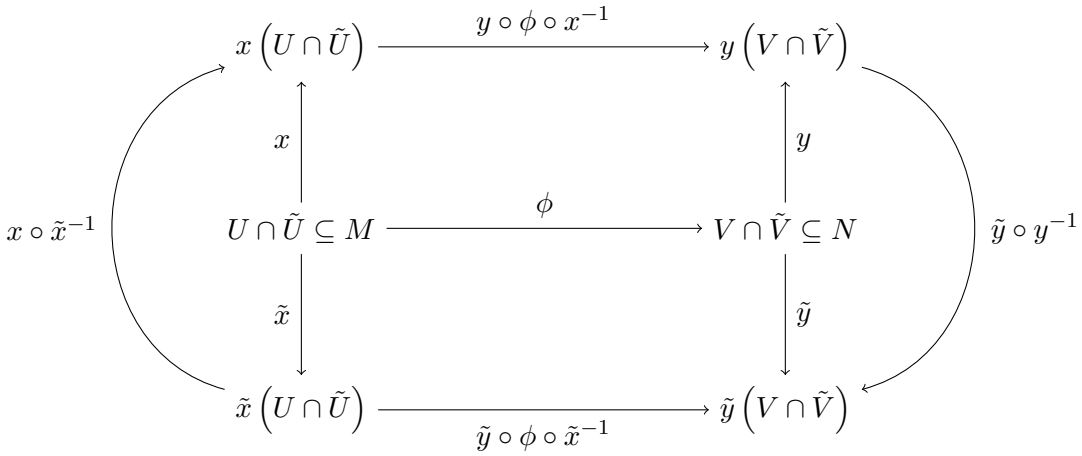
Let, $p \in U \subseteq M$, $\phi(p) \in V \subseteq N$, (U, x) and (V, y) both be charts.

ϕ is smooth at p iff $y \circ \phi \circ x^{-1}$ is smooth at $x(p) \in \mathbb{R}^d$.



This def^{n} of smoothness of ϕ does not care about the specific charts that we are picking. So it's a chart independent def^{n}.

Let, $p \in U \cap \tilde{U}$ and $\phi(p) \in V \cap \tilde{V}$



Now,

$$\tilde{y} \circ \phi \circ \tilde{x}^{-1} = (y \circ \tilde{y})^{-1} \circ \left(y \circ \phi \circ x^{-1}\right) \circ \left(x \circ \tilde{x}^{-1}\right)$$

$$\text{smooth} \impliedby \text{smooth} \quad \text{smooth} \quad \text{smooth}$$

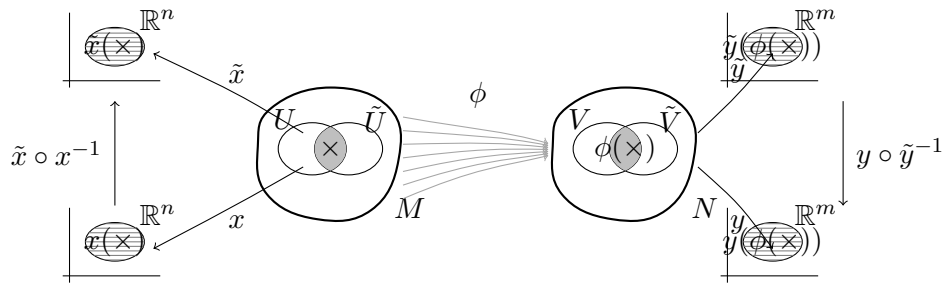


Fig: Amateur's depiction of a diffeomorphism ϕ

Definition 2.17. Diffeomorphism :=

$\phi : M \longrightarrow N$ be smooth map between smooth mfds. M and N. If $\phi^{-1} : N \longrightarrow M$ is also smooth, then we call ϕ a diffeomorphism. Diffeomorphism is the structure preserving map between smooth manifolds.

2.7 Algebra of Forms

At a point p in a d-dimensional manifold M, a special class of tensors are very important and deserves a seperate section of discussion. They are the totally antisymmetric (0,n)-tensors. To define them properly, let me first define an operator called the "Antisymmetrization Operator". For an (0,n)-tensor T at a point p in the manifold, and for a set of n linearly independent vectors $\{\overset{1}{X}, \overset{2}{X}, ..., \overset{n}{X}\}$ the action of the antisymmetrization operator $\hat{A}$ on the tensor T is defined as,

$$\hat{A}(T)\left(\overset{1}{X}, \overset{2}{X}, ..., \overset{n}{X}\right) = \frac{1}{n!}\sum_{\pi} \delta_\pi \, T\left(\overset{\pi(1)}{X}, \overset{\pi(2)}{X}, ..., \overset{\pi(n)}{X}\right)$$

where π denotes the different permutations of $\{1, 2, ..., n\}$ and δ_π is the sign of the permutation.

The defintion of $\hat{A}$ means, $\hat{A}\,(T)_{abc} = T_{[abc]}$, etc.

An n-form is a (0,n)-tensor which is the same as its antisymmetrized version. So, if $\hat{A}\,(T) = T$ then T is an n-form at p. As such the components of T in any chart is totally antisymmetric, i.e. antisymmetric under the exchange of any two pairs of indices. $T_{abc} = T_{[abc]}$.

It is obvious that there can not be an n-form where $n > d$. In that case, at least two of the indices must be the same. For example if $d = 3$, then a 4-form must have indices like T_{1233}. However, being totally antisymmetric, T_{1233} must be equal to $-T_{1233}$ and that is possible only if T_{1233} is 0.

For a component of an n-form to be non-zero, all the indices must be different. So,

for example, for a 2-form A in $d = 3$, $A_{11} = A_{22} = A_{33} = 0$, however, A_{12}, A_{13}, A_{23} can be non-zero.

We define a scalar as a 0-form. The space of all n-forms at the point p is denoted by $\Omega_p^n(M)$. The space of all n-form fields is denoted by $\Omega^n(M)$. An n-form field is also called a "differential n-form".

2.7.1 Wedge Product

Given a p-form A and a q-form B at the point p, we can make a $(p+q)$-form through an operation called wedge product. It is denoted by the symbol $\wedge$. Let us define this in stages. First, for a set of n 1-forms $\{\overset{1}{\eta}, ..., \overset{n}{\eta}\}$ their wedge product is defined as,

$$\overset{1}{\eta} \wedge ... \wedge \overset{n}{\eta} = n! \, \overset{[1}{\eta} \wedge ... \wedge \overset{n]}{\eta}$$

Suppose that at point p, we have a basis of T_p^*M, $\left\{E^1, ..., E^d\right\}$. The wedge product between any n of the basis elements is given by,

$$E^{a_1} \wedge ... \wedge E^{a_n} = n! \, E^{[a_1} \otimes ... \otimes E^{a_n]} \tag{2.2}$$

This, in turn, gives us a new way to express any $\omega \in \Omega_p^n(M)$ wrt the basis $\{E^i\}$.

$$\begin{aligned}
\omega &= \omega_{a_1 ... a_n} E^{a_1} \otimes ... \otimes E^{a_n} \\
&= \omega_{a_1 ... a_n} E^{[a_1} \otimes ... \otimes E^{a_n]} \\
&\text{The indices of } \omega \text{ are all totally antisymmetric.} \\
&= \frac{1}{n!} \omega_{a_1 ... a_n} \, n! \, E^{[a_1} \otimes ... \otimes E^{a_n]} \\
&\text{So, using eq. (2.2)} \\
\omega &= \frac{1}{n!} \omega_{a_1 ... a_n} E^{a_1} \wedge ... \wedge E^{a_n}
\end{aligned}$$

People also write it a bit differently sometimes, summing over only increasing indices or decreasing indices. finish this thought.

So now, given a p-form A and a q-form B, what are the components of $A \wedge B$ with respect to the basis $\left\{E^1, ..., E^d\right\}$?

$$\begin{aligned} A \wedge B &= \frac{1}{p!}\frac{1}{q!} A_{a_1 \ldots a_p} B_{b_1 \ldots b_q} E^{a_1} \wedge \ldots \wedge E^{a_p} \wedge E^{b_1} \wedge \ldots E^{b_q} \\ &= \frac{(p+q)!}{p!q!} A_{a_1 \ldots a_p} B_{b_1 \ldots b_q} E^{[a_1} \otimes \ldots \otimes E^{b_q]} \\ &= \frac{(p+q)!}{p!q!} A_{[a_1 \ldots a_p} B_{b_1 \ldots b_q]} E^{a_1} \otimes \ldots \otimes E^{b_q} \end{aligned}$$

So,

$$(A \wedge B)_{[a \ldots bc \ldots d]} = \frac{(p+q)!}{p!q!} A_{[a \ldots b} B_{c \ldots d]} \tag{2.3}$$

For two 1-forms ω and σ we know from the definition of one wedge products that $\omega \wedge \sigma = \omega \otimes \sigma - \sigma \otimes \omega$. This also means $\omega \wedge \sigma = -\sigma \wedge \omega$. The rule is, anytime we swap two one forms, there is a minus sign. Remembering this rule will lead us to a graded commutation rule for the wedge product of a p-form with a q-form. It is called graded, because the anti-commutation rule depends on the degree of the forms involved. Here is the rule (assuming that ω, σ are p and q forms, respectively).

$$\sigma \wedge \omega = (-1)^{pq}\, \omega \wedge \sigma \tag{2.4}$$

To see this, let's write the forms in a basis. So, $\omega = \frac{1}{p!}\omega_{a \ldots b} E^a \wedge \ldots \wedge E^b$ and $\sigma = \frac{1}{q!}\sigma_{c \ldots d} E^c \wedge \ldots \wedge E^d$. Now, $\sigma \wedge \omega = \frac{1}{p!q!}\sigma_{a \ldots b}\omega_{c \ldots d} E^a \wedge \ldots \wedge E^b \wedge E^c \wedge \ldots \wedge E^d$. To go to $\omega \wedge \sigma$, the trick is to take the basis vectors from the second set $E^c, \ldots, E^d$ one by one and take them p steps leftward. Each time I do this, I get a $(-1)^p$, and I have to do this for q members of the second set, so all in all, I have $(-1)^{pq}$. And in the end, the basis vectors will look like, $(-1)^{pq}\, E^c \wedge \ldots \wedge E^d \wedge E^a \wedge \ldots \wedge E^b$. So,

$$\begin{aligned} \sigma \wedge \omega &= \frac{1}{p!q!}\sigma_{a \ldots b}\omega_{c \ldots d} E^a \wedge \ldots \wedge E^b \wedge E^c \wedge \ldots \wedge E^d \\ &= (-1)^{pq}\, \frac{1}{p!q!}\sigma_{a \ldots b}\omega_{c \ldots d} E^c \wedge \ldots \wedge E^d \wedge E^a \wedge \ldots \wedge E^b \\ &= (-1)^{pq}\, \frac{1}{p!q!}\omega_{c \ldots d}\sigma_{a \ldots b} E^c \wedge \ldots \wedge E^d \wedge E^a \wedge \ldots \wedge E^b \\ &= (-1)^{pq}\, \omega \wedge \sigma \end{aligned}$$

2.7.2 Exterior Derivative

I will define this using coordinate basis. For derivatives, we need n-form fields as we can not do a derivation to a quantity which is defined at a point p. So we will work in a chart (U, x). The exterior derivative of a 0-form, i.e. a function f is defined as, $df := \partial_a f dx^a$. Notice that the exterior derivative of a zero form gives a 1-form. This is a general feature of exterior derivatives as we will see below that the exterior derivative of a p-form is a $(p+1)$-form. Now, given how a function's exterior derivative is taken, we define the exterior derivative of a p-form $A = \frac{1}{p!} A_{a_1 \ldots a_p} dx^{a_1} \wedge \ldots \wedge dx^{a_p}$ to be,

$$dA := \frac{1}{p!} d(A_{a_1 \ldots a_p}) \wedge dx^{a_1} \wedge \ldots \wedge dx^{a_p} \tag{2.5}$$

To see the components of dA, we expand the RHS of eq. (2.5). $dA = \frac{1}{p!} \partial_m A_{a_1 \ldots a_p} dx^m \wedge dx^{a_1} \wedge \ldots dx^{a_p} = \frac{(p+1)!}{p!} \partial_m A_{a_1 \ldots a_p} dx^{[m} \otimes dx^{a_1} \otimes \ldots \otimes dx^{a_p]} = (p+1)\, \partial_{[m} A_{a_1 \ldots a_p]} dx^m \otimes dx^{a_1} \otimes \ldots \otimes dx^{a_p}$. So the $dA_{m a_1 \ldots a_p}$ component of dA is given by $dA_{m a_1 \ldots a_p} = (p+1)\, \partial_{[m} A_{a_1 \ldots a_p]}$.

We reiterate here that the RHS of eq. (2.5) is a tensor by definition (exterior derivative of a function is a 1-form and wedge product of differential 1-forms is also a differential form).It is quite fascinating that a fully covariant notion of derivative can be cooked up from using only coordiante based partial derivative.

2.8 Integration on a Manifold

The following are very rough sketchy stuff. I was trying to do this and spent an awful lot of time without understanding much. Right now I am just keeping my stuff here and later on I will actually flesh them out.

2.8.1 Orientation and Volume Form

I will, for the time being, use η for volume forms.

ε will indicate Levi-Civita symbol.

Canonical n-form, i.e. the volume form,

$$\overset{M}{\eta} = \sqrt{|g|}\, \varepsilon$$

A clever way to write the Levi-Civita symbol is,

$$\varepsilon_{a_1 \ldots a_n} = n!\ \delta^1_{[a_1} \delta^2_{a_2} \delta^3_{a_3} \ldots \delta^n_{a_n]}$$

$$\overset{M}{\eta}_{a_1 \ldots a_n} = n! \sqrt{|g|}\ \delta^1_{[a_1} \ldots \delta^n_{a_n]}$$

$$\begin{aligned} \overset{M}{\eta}^{a_1 \ldots a_n} &= g^{a_1 b_1} \ldots g^{a_n b_n}\ \overset{M}{\eta}_{b_1 \ldots b_n} \\ &= g^{a_1 b_1} \ldots g^{a_n b_n}\ n! \sqrt{|g|}\ \delta^1_{[b_1} \ldots \delta^n_{b_n]} \\ &\text{(supply the argument here)} \\ &= (-1)^s\ n! \sqrt{|g|}\ \delta^{[a_1}_1 \ldots \delta^{a_n]}_n \end{aligned}$$

Again, supply $\longrightarrow$

$$\overset{M}{\eta}^{ab \ldots d}\ \overset{M}{\eta}_{ef \ldots h} = (-1)^s\ n!\ \delta^a_{[e} \delta^b_f \ldots \delta^d_{h]}$$

$$\overset{M}{\eta}^{ab \ldots d}\ \overset{M}{\eta}_{ab \ldots d} = (-1)^s\ n!\,\text{(supply the const.)}$$

$$\nabla_m \overset{M}{\eta}^{a \ldots b} = 0 = \nabla_m \overset{M}{\eta}_{a \ldots b} \qquad \text{(show)}$$

Say I have a vector X^a.
Define an $(n-1)$-form α by,

$$\alpha_{a_1\ldots a_{n-1}} = \overset{M}{\eta}_{ba_1\ldots a_{n-1}} v^b$$

$$\begin{aligned}(d\alpha)_{ca_1\ldots a_{n-1}} &= n\nabla_{[c}\alpha_{a_1\ldots a_{n-1}]} \\ &= n\nabla_{[c}\left(\overset{M}{\eta}_{|b|a_1\ldots a_{n-1}]} v^b\right) \\ &= n\overset{M}{\eta}_{b[a_1\ldots a_{n-1}}\nabla_{c]} v^b\end{aligned}$$

Also,

$$\begin{aligned}&(d\alpha)_{ca_1\ldots a_{n-1}} = h\overset{M}{\eta}_{ca_1\ldots a_{n-1}} \\ &\implies n\overset{M}{\eta}_{b[a_1\ldots a_{n-1}}\nabla_{c]} v^b = h\overset{M}{\eta}_{ca_1\ldots a_{n-1}}\end{aligned}$$

Again show this yourself. How to go from (B.2.20) to (B.2.21)
Finally, $d\alpha = (\nabla_a v^a)\overset{M}{\eta}$
Stokes $\implies$

$$\int_V (\nabla_a v^a)\overset{M}{\eta} = \int_{\partial V}\alpha := \int_{\partial V}\phi^*(\bar{\alpha})$$

Here $\bar{\alpha}$ = the $(n-1)$-form α restricted to act only on vectors $X \in \mathcal{T}$.
Now, let me take the following Gaussian Normal coordinates.

2.9 Non-coordinate Basis and Tetrad Formalism

For non-coordinate basis, I will use Greek indices and for coordinate basis I will use Latin indices.

Here is a consistent story,

$$g\left(E_\mu, E_\nu\right) = \eta_{\mu\nu} \tag{2.6}$$

Since E^μ is a covecotor, it can be expanded in terms of the coordinate basis,

$$E^\mu = E^\mu{}_a dx^a$$

Now, E can be thought of as a matrix. Our convention is as usual, upper index is for row, lower index is for column. And, for any matrix A, $(A^{-1})^i{}_j$ is denoted by $A_j{}^i$. Based on this, dx^a can be expanded as,

$$dx^a = E_\mu{}^a E^\mu$$

Now, the non-coordinate basis vectors E_μ are dual to E^μ, that is, $E^\mu(E_\nu) = \delta^\mu_\nu$. The basis vectors E_μ can also be expanded in terms of the coordiante basis vectors ∂_a, as, $E_\mu = F_\mu{}^a \partial_a$. The nice thing is that F is nothing but E! Let me show that.

$$\begin{aligned}
&E^\mu(E_\nu) = \delta^\mu_\nu \\
&E^\mu{}_a dx^a \left(F_\nu{}^b \partial_b\right) = \delta^\mu_\nu \\
&E^\mu{}_a F_\nu{}^b dx^a(\partial_b) = \delta^\mu_\nu \\
&E^\mu{}_a F_\nu{}^b \delta^a_b = \delta^\mu_\nu \\
&E^\mu{}_a F_\nu{}^a = \delta^\mu_\nu \\
&E^\mu{}_a \left(F^{-1}\right)^a{}_\nu = \delta^\mu_\nu
\end{aligned}$$

This simply means, $F^{-1} = E^{-1}$ or, $F = E$. And,

$$E_\mu = E_\mu{}^a \partial_a$$

It is now easy to see how ∂_a is expressed as a linear combination of E_μ.

$$\begin{aligned}
&E_\mu = E_\mu{}^a \partial_a \\
&E_\mu E^\mu{}_b = E^\mu{}_b E_\mu{}^a \partial_a \\
&E^\mu{}_b E_\mu = \delta^a_b \partial_a \\
&\partial_b = E^\mu{}_b E_\mu \\
&\partial_a = E^\mu{}_a E_\mu
\end{aligned}$$

To summarize, we have the following rules,

- ${E^\mu}_a$ is the a-th component of the covector E^μ in the coordiante basis $\{dx^a\}$ of covectors or the μ-th component of the vector ∂_a in the frame basis $\{E_\mu\}$ of vectors.

$$E^\mu = {E^\mu}_a dx^a$$
$$\partial_a = {E^\mu}_a E_\mu$$

- ${E_\mu}^a$ is the a-th component of the vector E_μ in the coordiante basis $\{\partial_a\}$ of vectors or the μ-th component of the covector dx^a in the frame basis $\{E^\mu\}$ of covectors.

$$E_\mu = {E_\mu}^a \partial_a$$
$$dx^a = {E_\mu}^a E^\mu$$

- We always put the Greek index closer to the letter E and the Latin index further from the letter E.

It is nice to see how handy the tensor index gymnastics can be. It is almost like a magic. I can go back and forth between a tensor with its compoenet/s taken in the coordinate basis or the frame basis. For example, say I have a (2,0) tensor A. Then, $A^{\mu m} = A^{nm}{E^\mu}_n$, $A^{m\mu} = A^{\alpha\mu}{E_\alpha}^m$ etc. We can easily show these two, and we hope this is enough to convince a normal person about the general validity of this method.

$$\begin{aligned} A^{\mu m} &= A\left(E^\mu, dx^m\right) \\ &= A\left({E^\mu}_n dx^n, dx^m\right) \\ &= {E^\mu}_n A\left(dx^n, dx^m\right) \\ &= {E^\mu}_n A^{nm} \end{aligned}$$

and , for the second equality,

$$\begin{aligned} A^{m\mu} &= A\left(dx^m, E^\mu\right) \\ &= A\left(E_\nu{}^m E^\nu, E^\mu\right) \\ &= \end{aligned}$$

2.9.1 Connection 1-form and Ricci Rotation Coefficient

Since E_μ is a field (at least in a chart U), I can take the covariant derivative of it wrt the basis vector fields of that chart. The result will have to be another vector field and can be expanded in terms of $\{E_\mu\}$.

$$\nabla_{\partial_a} E_\mu = \omega^\nu{}_{\mu a} E_\nu$$

We can show that although $\omega^\nu{}_{\mu a}$ is a 1-form in the index a, it is not a tensor in the indices μ and ν. First, notice that the definition of $\omega^\nu{}_{\mu a}$ can be given as, $\omega^\nu{}_{\mu a} = E^\nu\left(\nabla_{\partial_a} E_\mu\right)$. Now, say I change the chart, but not the frame $\{E_\mu\}$. So,

$$\begin{aligned} \omega^\nu{}_{\mu a'} =& E^\nu\left(\nabla_{\partial'_a} E_\mu\right) \\ =& E^\nu\left(\nabla_{\left(\frac{\partial x^m}{\partial x'^a}\partial_m\right)} E_\mu\right) \\ =& E^\nu\left(\frac{\partial x^m}{\partial x'^a}\nabla_{\partial_m} E_\mu\right) \\ =& \frac{\partial x^m}{\partial x'^a} E^\nu\left(\nabla_{\partial_m} E_\mu\right) \\ =& \frac{\partial x^m}{\partial x'^a}\omega^\nu{}_{\mu m} \end{aligned}$$

This proves that $\omega^\nu{}_{\mu a}$ is a 1-form in the index a. To show that it is NOT a tensor in the Greek indices, I now change only the frame and not the chart, then,

$$\begin{aligned} \omega^{\nu'}{}_{\mu' a} &= E'^\nu\left(\nabla_{\partial_a} E'_\mu\right) \\ &= \Lambda^\nu{}_\alpha E^\alpha\left(\nabla_{\partial_a}\left(\Lambda_\mu{}^\beta E_\beta\right)\right) \\ &= \Lambda^\nu{}_\alpha E^\alpha\left(\Lambda_\mu{}^\beta \nabla_a E_\beta + E_\beta \nabla_a \Lambda_\mu{}^\beta\right) \\ &= \Lambda^\nu{}_\alpha E^\alpha\left(\Lambda_\mu{}^\beta \omega^\rho{}_{\beta a} E_\rho + E_\beta \nabla_a \Lambda_\mu{}^\beta\right) \\ &= \Lambda^\nu{}_\alpha \Lambda_\mu{}^\beta \omega^\rho{}_{\beta a}\delta^\alpha_\rho + \Lambda^\nu{}_\alpha \partial_a \Lambda_\mu{}^\beta \delta^\alpha_\beta \end{aligned}$$

$$= \Lambda^{\nu}{}_{\alpha}\Lambda_{\mu}{}^{\beta}\omega^{\alpha}{}_{\beta a} + \Lambda^{\nu}{}_{\alpha}\partial_a\Lambda_{\mu}{}^{\alpha}$$

The last term shows the inhomogenous nature of the transformation of $\omega^{\nu}{}_{\mu a}$ under LT of the frame.

Write down expressions for $\nabla_\nu A^\mu$ just like you know how to write $\nabla_m A^a$ etc.

Now, just as we know how to expand $\nabla_b A^a$ in terms of a partial derivative and Christoffel symbol, we can derive a similar expression for $\nabla_m A^\mu$. To do so, notice that $\nabla_m A^\mu$ is defined as $E^\mu\left(\nabla_{\partial_m} A\right) = E^\mu\left(\nabla_{\partial_m}\left(A^\nu E_\nu\right)\right) = E^\mu\left(E_\nu \partial_m A^\nu + A^\nu \nabla_{\partial_m} E_\nu\right) = E^\mu\left(E_\nu \partial_m A^\nu + A^\nu \omega^{\rho}{}_{\nu m} E_\rho\right) = \partial_m A^\mu + \omega^{\mu}{}_{\nu m} A^\nu$. So,

$$\nabla_m A^\mu = \partial_m A^\mu + \omega^{\mu}{}_{\nu m} A^\nu$$

One immediate inquiry is the expression of $\omega^{\nu}{}_{\mu a}$ in terms of Γ^a_{bc}. We know that,

$$\begin{aligned}
&\nabla_a A^\mu = \left(\nabla_a A^b\right) E^{\mu}{}_{b} \\
&\partial_a\left(A^\mu\right) + \omega^{\mu}{}_{\rho a} A^\rho = \left(\partial_a A^b + \Gamma^b_{ma} A^m\right) E^{\mu}{}_{b} \\
&\partial_a\left(E^{\mu}{}_{m} A^m\right) + \omega^{\mu}{}_{\rho a} A^\rho = \left(\partial_a A^b + \Gamma^b_{ma} A^m\right) E^{\mu}{}_{b} \\
&\partial_a\left(E^{\mu}{}_{m} A^m\right) + \omega^{\mu}{}_{\rho a} E^{\rho}{}_{m} A^m = \left(\partial_a A^b + \Gamma^b_{ma} A^m\right) E^{\mu}{}_{b} \\
&A^m \partial_a E^{\mu}{}_{m} + E^{\mu}{}_{m}\partial_a A^m + \omega^{\mu}{}_{\rho a} E^{\rho}{}_{m} A^m = E^{\mu}{}_{m}\partial_a A^m + \Gamma^n_{ma} E^{\mu}{}_{n} A^m \\
&A^m \partial_a E^{\mu}{}_{m} + \omega^{\mu}{}_{\rho a} E^{\rho}{}_{m} A^m = \Gamma^n_{ma} E^{\mu}{}_{n} A^m \\
&\partial_a E^{\mu}{}_{m} + \omega^{\mu}{}_{\rho a} E^{\rho}{}_{m} = \Gamma^n_{ma} E^{\mu}{}_{n} \\
&\text{Now simply multiply by } E_\nu{}^m \text{ on both sides.} \\
&E_\nu{}^m \partial_a E^{\mu}{}_{m} + \omega^{\mu}{}_{\rho a} E^{\rho}{}_{m} E_\nu{}^m = \Gamma^n_{ma} E^{\mu}{}_{n} E_\nu{}^m \\
&E_\nu{}^m \partial_a E^{\mu}{}_{m} + \omega^{\mu}{}_{\rho a}\delta^\rho_\nu = \Gamma^n_{ma} E^{\mu}{}_{n} E_\nu{}^m \\
&\omega^{\mu}{}_{\nu a} = \Gamma^n_{ma} E^{\mu}{}_{n} E_\nu{}^m - \left(\partial_a E^{\mu}{}_{m}\right) E_\nu{}^m
\end{aligned} \tag{2.7}$$

Similarly, multiplying both sides of eq. (2.7) with $E_\mu{}^p E^\nu{}_q$ so that the two E's with the Christoffel symbol becomes Kronecker deltas and ultimately go away,

$$\begin{aligned}
&\omega^{\mu}{}_{\nu a} E_\mu{}^p E^\nu{}_q = \Gamma^n_{ma} E_\mu{}^p E^{\mu}{}_{n} E_\nu{}^m E^\nu{}_q - \partial_a E^{\mu}{}_{m}\; E_\nu{}^m E^\nu{}_q E_\mu{}^p \\
&\omega^{\mu}{}_{\nu a} E_\mu{}^p E^\nu{}_q = \Gamma^n_{ma}\delta^p_n\delta^m_q - \partial_a E^{\mu}{}_{m}\delta^m_q E_\mu{}^p
\end{aligned}$$

$$\omega^{\mu}{}_{\nu a} E_{\mu}{}^{p} E^{\nu}{}_{q} = \Gamma^{p}_{qa} - \partial_a E^{\mu}{}_{q}\ E_{\mu}{}^{p}$$
$$\Gamma^{p}_{qa} = \omega^{\mu}{}_{\nu a} E_{\mu}{}^{p} E^{\nu}{}_{q} + \partial_a E^{\mu}{}_{q}\ E_{\mu}{}^{p}$$
Just replace p with m, q with n, and swap μ and ν
$$\Gamma^{m}_{na} = \omega^{\nu}{}_{\mu a} E_{\nu}{}^{m} E^{\mu}{}_{n} + \left(\partial_a E^{\nu}{}_{n}\right) E_{\nu}{}^{m} \tag{2.8}$$

This leaves us to defining another quantity in this section, called the Ricci rotation coefficients. It is basically the connection one form expressed in the frame basis instead of coordinate basis. That is, while a connection one form $\omega^{\nu}{}_{\mu a}$ has the 1-form index "a" from the coordinate basis, the Ricci rotation coefficient $\omega^{\nu}{}_{\mu\rho}$ is the connection 1-form expanded in terms of $\{E^{\mu}\}$.

$$\omega^{\nu}{}_{\mu\rho} = \omega^{\nu}{}_{\mu a} E_{\rho}{}^{a} \tag{2.9}$$

Another way to express the Ricci rotation coefficient is, $\omega^{\nu}{}_{\mu\rho} = E^{\nu}\left(\nabla_{E_\rho} E_\mu\right)$. We can quickly see that as follows, by starting with the RHS,

$$\begin{aligned} E^{\nu}\left(\nabla_{E_\rho} E_\mu\right) &= E^{\nu}\left(\nabla_{(E_\rho{}^{m}\partial_m)} E_\mu\right) \\ &= E^{\nu}\left(E_\rho{}^{m} \nabla_{\partial_m} E_\mu\right) \\ &= E_\rho{}^{m} E^{\nu}\left(\nabla_{\partial_m} E_\mu\right) \\ &= E_\rho{}^{m} \omega^{\nu}{}_{\mu m} \\ &= \omega^{\nu}{}_{\mu\rho} \end{aligned}$$

Properties of Ricci Rotation Coefficients

Several properties of Ricci Rotation coefficients are important to understand.

2.9.2 Tensor Valued Forms and Exterior Derivatives

Torsion and Curvature 2-forms

I will first simply define the Torsion and Curvature 2-forms, then show that these definitions work. There are other ways of arriving at these definitions that might be more direct, but since we have to memorize these anyways, I would simply state the definition without motivation.

The torsion 2-form is defined as (suppressing the latin indices),

$$T^{\mu} = dE^{\mu} + \omega^{\mu}{}_{\nu} \wedge E^{\nu} \tag{2.10}$$

The curvature 2-form is defined as (again, supressing the latin indices),

$$R^{\mu}{}_{\nu} = d\omega^{\mu}{}_{\nu} + \omega^{\mu}{}_{\alpha} \wedge \omega^{\alpha}{}_{\nu} \tag{2.11}$$

To show that the definition of torsion form works, start with expanding the RHS of eq. (2.10),

$$dE^{\mu}{}_{ab} + \left(\omega^{\mu}{}_{\nu} \wedge E^{\nu}\right)_{ab} = \partial_a E^{\mu}{}_{b} - \partial_b E^{\mu}{}_{a} + \omega^{\mu}{}_{\nu a} E^{\nu}{}_{b} - \omega^{\mu}{}_{\nu b} E^{\nu}{}_{a} \tag{2.12}$$

Now,

$$\begin{aligned}
T^{\mu}{}_{ab} &= T\left(E^{\mu}, \partial_a, \partial_b\right) \\
&= E^{\mu}\left(\nabla_a \partial_b - \nabla_b \partial_a\right) \\
&= E^{\mu}\left(\nabla_a \partial_b\right) - (a \leftrightarrow b) \\
&= E^{\mu}\left(\Gamma^{m}_{ba} \partial_m\right) - (a \leftrightarrow b) \\
&= E^{\mu}\left(\Gamma^{m}_{ba} E^{\nu}{}_{m} E_{\nu}\right) - (a \leftrightarrow b) \\
&= \Gamma^{m}_{ba} E^{\nu}{}_{m} \delta^{\mu}_{\nu} - (a \leftrightarrow b) \\
&= \Gamma^{m}_{ba} E^{\mu}{}_{m} - (a \leftrightarrow b) \\
&= E^{\mu}{}_{m}\left(\omega^{\beta}{}_{\alpha a} E_{\beta}{}^{m} E^{\alpha}{}_{b} + \partial_a E^{\beta}{}_{b}\, E_{\beta}{}^{m}\right) - (a \leftrightarrow b) \\
&= \omega^{\beta}{}_{\alpha a} E^{\mu}{}_{m} E_{\beta}{}^{m} E^{\alpha}{}_{b} + \partial_a E^{\beta}{}_{b}\, E^{\mu}{}_{m} E_{\beta}{}^{m} - (a \leftrightarrow b) \\
&= \omega^{\beta}{}_{\alpha a} \delta^{\mu}_{\beta} E^{\alpha}{}_{b} + \partial_a E^{\beta}{}_{b}\, \delta^{\mu}_{\beta} - (a \leftrightarrow b) \\
&= \partial_a E^{\mu}{}_{b} + \omega^{\mu}{}_{\alpha a} E^{\alpha}{}_{b} - (a \leftrightarrow b) \\
&= \partial_a E^{\mu}{}_{b} + \omega^{\mu}{}_{\nu a} E^{\nu}{}_{b} - \partial_b E^{\mu}{}_{a} - \omega^{\mu}{}_{\nu b} E^{\nu}{}_{a} \\
&= \partial_a E^{\mu}{}_{b} - \partial_b E^{\mu}{}_{a} + \omega^{\mu}{}_{\nu a} E^{\nu}{}_{b} - \omega^{\mu}{}_{\nu b} E^{\nu}{}_{a}
\end{aligned}$$

This matches with eq. (2.12)

Chapter 3

Derivatives on Manifolds

3.1 Introduction

In this chapter, we use some of the concepts introduced in the previous one to talk about the very fascinating concept of differentiation on a manifold. We all have an idea about what a differentiation is. It is a small change in something over small change in some parameter value. For example, the x component of velocity is the change in position on the x axis, Δx over change in the parameter time, Δt. When taking a derivative of a tensor field over a manifold, we need to carefully define what change in the tensor is and what parameter comes in the denominator. It turns out that there are at least three interesting ways we can define a change in a tensor, and they will give rise to three different types of derivatives.

Now the naive way of taking the change of a tensor T between nearby points P and Q is to simply take the difference, T(Q)-T(P). This does not work for the simple reason that P and Q are different points, even if they are close, and as such, the tensors T(Q) and T(P) simply live in completely different vector spaces. It just is not possible to take their difference.

This gives rise to the need to introduce a connecting mechanism that will connect the tensor at T(Q) and give a unique counterpart of T(Q) at the point P, call it T$'$(P), which belongs to the same vector space as T(P). Now we can define the difference as T$'$(P)-T(P).

3.2 Lie Derivative

Here, the connecting mechanism comes from a smooth vector field, X. The local 1 parameter group of diffeomorphism generated by X introduce flow lines. The two points p and q will lie close together and on the same flow line.

This is how the definition goes,

Definition 3.1 (Lie Derivative)**.** Let T be a tensor field over M and p be a point in M. Let X be a vector field and $\{\Gamma_t : U \longrightarrow \Gamma_t[U]\}_{t\in I}$ be a local 1 parameter family of diffeomorphism generated by X, where U is an open subset of M and p$\in$ U, and I is an open subset of $\mathbb{R}$. Then the Lie derivative of T with respect to X, at the point p is defined as,

$$\pounds_X T = \lim_{t\to 0} \frac{(\Gamma_t)^* T - T}{t}$$

I have avoided writing $\Big|_p$ on all of the tensors to avoid clutter, and I will consistently do this kind of abuse of notation but it will be clear from context.

The right hand side of the definition tells us how the mechanism of connecting back. To evaluate the Lie derivative of T wrt X at the point p, I first go parameter distance t away from the point p along the maximal integral curve of X with initial point p, that is, at the point $\Gamma_t(p)$. There, the tensor $\mathrm{T}\Big|_{\Gamma_t(p)}$ is taken. Then using the pull-back map $(\Gamma_t)^*$ I pull that tensor back to the point p, making $(\Gamma_t)^* T\Big|_p$. Then I subtract from this the tensor $\mathrm{T}\Big|_p$. This is how the difference is defined. And the parameter in the denominator is simply the parameter distance t along the maximal integral curve of X with initial point p.

Proposition 3.2. The operator $\pounds_X$ has the following properties.

1. Commutes with addition. $\pounds_X(T+S) = \pounds_X T + \pounds_X S$
2. Satisfies the Leibniz rule. $\pounds_X(T\otimes S) = (\pounds_X T)\otimes S + T\otimes(\pounds_X S)$
3. Commutes with index substitution and contraction

3.3 Covariant Derivative

A **connection** ∇ on a smooth manifold $(M, \mathcal{O}, \mathcal{A})$ is a map that takes a pair consisting of a vector (field) X and a (p,q)-tensor field T and sends them to a (p,q)-tensor field $\nabla_X T$ satisfying –

(D1) $\nabla_{(fX+Z)}T = f\nabla_X T + \nabla_Z T$

(D2) $\nabla_X(aT+bS) = a\nabla_X T + b\nabla_X S \quad (a,b\in\mathbb{R})$

(D3) Leibniz: $\nabla_X(T\otimes S) = T\otimes\nabla_X S + \nabla_X T\otimes S$

(D4) $\nabla_X f = X(f)$

The reason to use brackets around the word “field” is that X can be a defined only at a point too.

Chapter 4

Submanifolds and Hypersurfaces

4.1 Introduction

This is a crucial ingredient in understanding almost anything that has to do with Numerical Relativity, the art of solving EFEs by the help of numerical methods. I will closely follow the treatment given by David Malament in his book on the foundations of GR and Newtonian gravity. In my opinion, among all the GR textbooks, this one has the most systematic treatment of this topic. Though, being written by a philosopher, the main idea of the book is not to teach differential geometry so that one can use it in research, rather it is to present only that much which forms the core of our understanding of the topic. However, this much is actually sufficient for diving deep into a proper numerical relativity textbook.

4.2 Submanifolds

We will discuss the idea of a submanifold first. Hypersurfaces are a special tyrpe of submanifold. The key idea needed to be introduced to define a submanifold is the idea of imbedding.

4.2.1 Imbedding

Let S and M be two manifolds. Let there be a smooth map $\Psi : S \longrightarrow M$. Ψ is called an imbedding if the following three conditions hold.

- Ψ is injective.
- at all p in S, the push forward map $(\Psi_p)_* : T_pS \longrightarrow T_pM$ is injective.
- the inverse map $\Psi^{-1} : \Psi[S] \longrightarrow S$ is continuous with respect to the relative topology in $\Psi[S]$.

At first glance, this definition might seem unmotivated and built out of thin air. TODO: learn and write the reasons behind this definition

Now, suppose that S⊆M, then we say that S is an imbedded submanifold of M if the map $id : S \longrightarrow M$ is an imbedding. If the dimension of M is n, and the dimension of S is k, and $k = n - 1$ then we say that S is a *hypersurface* in M.

In this and the next few sections, let me fix S to be a k-dimensional imbedded submanifold of M. Later on we will think more specifically in terms of hypersurfaces. But for now let me go through some mathematical machinary needed for imbedded submanifolds in general.

4.2.2 Tensors Tangent and Normal to S

At a point p in M, we can talk about tensors with respect to M, or with respect to S. We refer to the former as M-tensors, and the later as S-tensors. The S-tensors are S's own objects and we do not need to refer to M while talking about them. We will use a tilde over the name of a tensor to show that it is an S-tensor.

If S is a metric submanifold of M (I will explain what a metric submanifold is in a bit), we can divide T_pM into two subspaces, one is a subspace of vectors "tangent" to S, and the other one is a subspace of tensors "normal" to S. (I am using quotes here because this "tangent" is not related to the word "tangent" in "tangent vector").

Definition 4.1. A vector $X \in T_pM$ is *tangent* to S if $X \in (id_p)_*(T_pS)$.

Definition 4.2. A covector $\eta \in T_p^*M$ is *normal* to S if for all vectors $X \in T_pM$ that are tangent to S, $\eta(X) = 0$

We denote the space of all M-vectors tangent to S at p as $\mathscr{T}_p$ and the space of all M-covectors normal to S at p as $\mathscr{N}_p^*$. As we took T_pS and then pushed it forward with an imbedding (which basically maps T_pS to T_pM injectively), the dimension of $\mathscr{T}_p$ is k. We can also show that the dimension of $\mathscr{N}_p^*$ is $n - k$.

Another way to describe tangent vector comes from what it means to push forward a vector from one manifold to another. For a vector $\tilde{X}$ in S, there is obviously a curve γ to which the vector $\tilde{X}$ is tangent to. Since push forward of a vector tangent to a curve is tangent to the image of the curve, $id_*\tilde{X}$ is tangent to $id(\gamma)$. In fact, we are considering S to be a subset of M, and $id(\gamma)$ is actually just γ. This means, any M-vector at p that is tangent to a curve γ which lies entirely in S, is a memeber of $\mathscr{T}_p$. Wait, how do you show the converse.....?????

Proposition 4.3. The dimension of $\mathscr{N}_p^*$ is $n - k$.

Proof. First of all, we know that dimension of $\mathscr{T}_p$ is k. Take k vectors $\overset{1}{X}, \overset{2}{X}, ..., \overset{k}{X}$ in $\mathscr{T}_p$ that are linearly independent. Now, to make a basis of T_pM we need some

other linearly independent vectors outside of $\mathscr{T}_p$. Let's say the total basis is $\{\overset{1}{X}, \overset{2}{X}, ..., \overset{k}{X}, \overset{k+1}{X}, ..., \overset{n}{X}\}$. Let us take a basis of the space of covectors at p, T_p^*M that is dual to the basis of vectors that we have just taken : $\{\overset{1}{\eta}, \overset{2}{\eta}, ..., \overset{k}{\eta}, \overset{k+1}{\eta}, ..., \overset{n}{\eta}\}$. Then, the covectors $\overset{k+1}{\eta}, ..., \overset{n}{\eta}$ span a space $\mathscr{N}'$ with the following property: any member η of $\mathscr{N}'$, being a combination of $\overset{k+1}{\eta}, ..., \overset{n}{\eta}$, will actually produce $\eta(X) = 0$ whenever acting on any vector $X \in \mathscr{T}_p$ since X is a combination of $\overset{1}{X}, \overset{2}{X}, ..., \overset{k}{X}$. Evidently, the space $\mathscr{N}'$ is actually what we are calling $\mathscr{N}_p^*$. And the dimension is $n - k$ □

Once we have $\mathscr{N}_p^*$ defined, we can prove the following proposition, which will be very useful later.

Proposition 4.4. An M-tensor X is tangent to S at p iff for any $\eta \in \mathscr{N}_p^*$, $\eta(X) = 0$.

Proof. Let us use the basis vectors as defined before. T_pM is spanned by $\{\overset{1}{X}, \overset{2}{X}, ..., \overset{n}{X}\}$ where the first k vectors span $\mathscr{T}_p$. Let T_p^*M be spanned by the dual vectors $\{\overset{1}{\eta}, \overset{2}{\eta}, ..., \overset{n}{\eta}\}$ where the last $n - k$ covectors span $\mathscr{N}_p^*$.

Now, let $X \in \mathscr{T}_p$. By our definition of the space of covectors at p that are normal to S, any member η of $\mathscr{N}_p^*$ will follow $\eta(X) = 0$.

Lets now assume that we have an $X \in T_pM$ such that for any $\eta \in \mathscr{N}_p^*$, $\eta(X) = 0$. This means in the expansion of X, there cannot be any term of the form $a_j \overset{j}{X}$ where $j > k$. And so X is a linear combination of only $\{\overset{1}{X}, ...\overset{k}{X}\}$. And that means, $X \in \mathscr{T}_p$ □

While the M-vectors tangent to S come directly from S via a push forward map, the M-covectors pulled back to S actually give zero. This is expected, as they are "normal" to S.

Proposition 4.5. $\eta \in \mathscr{N}_p^* \iff (id_p)^*(\eta) = 0$

Proof. First assume that $\eta \in \mathscr{N}_p^*$.

Let $\tilde{X} \in T_pS$. So, $((id_p)^*(\eta))\left(\tilde{X}\right) = \eta\left((id_p)_*\tilde{X}\right) = 0$. Since this is true for any $\tilde{X}$ in T_pS, we can conclude that $(id_p)^*(\eta) = 0$.

Next assume that $(id_p)^*(\eta) = 0$ where $\eta \in T_p^*M$.

For any $\tilde{X} \in T_pS$, $((id_p)^*(\eta))(\tilde{X}) = 0 \implies \eta\left((id_p)_*(\tilde{X})\right) = 0$. This means η acting on any member of $\mathscr{T}_p$ is zero. Hence, $\eta \in \mathscr{N}_p^*$.

□

Now, more general M-tensors at p can also be tangent and normal, in different indices. For example, we can say that a (2,3) M-tensor $T^{ab}{}_{cde}$ at p is tangent in the first upper index a, or normal in the third lower index e. The follwing definition can easily be generalized.

Definition 4.6. A (2,3) M-tensor $T^{ab}{}_{cde}$ at point p is tangent to S in the second upper index b if for any $\eta \in \mathscr{N}_p^*$, $T^{ab}{}_{cde}\eta_b = 0$. It is normal to S in the third lower index e if for any $X \in \mathscr{T}_p$, $T^{ab}{}_{cdn}X^n = 0$. And so on for tensors of different index structures.

So far we have only considered being tangent on an upper index, and being normal on a lower index. But it is also possible to be normal on an upper index and to be tangent on a lower index if the manifold M is a metric manifold. Even at this point I do not need the submanifold to have a metric.

Definition 4.7. $T^{ab}{}_{cde}$ is normal in the first upper index a if $T_a{}^b{}_{cde} = g_{am}T^{mb}{}_{cde}$ is normal in the index a. Similarly $T^{ab}{}_{cde}$ is tangent in the second lower index d if $T^{ab}{}_c{}^d{}_e = g^{dm}T^{ab}{}_{cme}$ is tangent in the index d. And so on for tensors of other index structure.

Basically, if after raising a lower index the tensor is tangent in that raised index, then it is tangent in that index in its lowered form too. And if after lowering an index, the tensor is normal in that index, then the tensor is normal in that index in the raised form too.

Right now, for M-vectors and M-covectors at p, we have 4 different spaces in terms of being tangent or being normal to S.

- $\mathscr{T}_p$ is the space of M-vectors tangent to S at p.
- $\mathscr{T}_p^*$ is the space of M-covectors tangent to S at p.
- $\mathscr{N}_p$ is the space of M-vectors normal to S at p.
- $\mathscr{N}_p^*$ is the space of M-covectors normal to S at p.

We know that the dimension of $\mathscr{T}_p$ is k and the dimension of $\mathscr{N}_p^*$ is $(n-k)$. What is the dimension of $\mathscr{T}_p^*$ and $\mathscr{N}_p$?

Proposition 4.8. The dimesion of $\mathscr{T}_p^*$ is k and the dimension of $\mathscr{N}_p$ is $(n-k)$.

Proof. We know that the metric tensor g_{ab} creates a bijective mapping between T_pM and T_p^*M. So thinking from that perspective, $\mathscr{T}_p^*$ is basically the set of all covectors η such that $\eta_a = g_{am}X^m$ where $X \in \mathscr{T}_p$. And the dimension of such a space is obviously the dimension of $\mathscr{T}_p$, that is, k.

In a similar way, the space $\mathscr{N}_p$ is made of all vectors X that has the form $X^a = g^{am}\eta_m$ where $\eta \in \mathscr{N}_p^*$. And the dimension of such a space is equal to the dimension of $\mathscr{N}_p^*$, which is $(n-k)$. □

A key fact in this case is that for metric manifolds, at a point p, $T_pM = \mathscr{T}_p \oplus \mathscr{N}_p$. Elaborate on this and prove

4.2.3 Sticking to Only M-Tensors

At any point p, there is a natural isomorphism between S-tensors of any index structure and M-tensors of the same index structure that are tangent to S. This is the very basis of how we talk about a hypersurface. The idea is that we will commit to M-tenosrs tangent to S and not talk about S-tensors.

Let me first describe the isomorphism. Here I will satisfy myself by only defining the map, and I will skip the proof of the map being an isomorphism.

The map ϕ_p takes an S-tensor $\tilde{X}^{ab}{}_{cd}$ to $\phi_p\left(\tilde{X}^{ab}{}_{cd}\right)$, which is an M-tensor. We define the isomorphism in stages, first for scalars, then for vectors, covectors and finally for tensors of arbitray index structure.

First, for a scalar α at p, $\phi_p(\alpha) = \alpha$. There is no need to use tilde as scalars are just numbers.

Second, for an S-vector $\tilde{X}^a$ at p, $\phi_p\left(\tilde{X}^a\right) = (id_p)_* \tilde{X}^a$. By the very definition of a vector being tangent to S at p, $\phi_p\left(\tilde{X}^a\right)$ is tangent to S.

Third, for a S-covector $\tilde{\eta}_a$ at p, $\phi_p(\tilde{\eta}_a)$ is defined by its action on any M-vector X^a.

$$\phi_p(\tilde{\eta}_a) X^a = \begin{cases} \tilde{\eta}_a\left(\phi_p^{-1} X^a\right); & X \in \mathscr{T}_p \\ 0; & X \in \mathscr{N}_p \end{cases}$$

We remark that this is well defined, as $\phi_p^{-1} X^a$ is well defined. Also, notice that the second clause of the definition of $\phi_p(\tilde{\eta}_a)$ ensures that it is tangent to S.

Lastly, lets take an arbitray index structure S-tensor $\tilde{X}^{ab}{}_c$, then,

$$\phi_p\left(\tilde{X}^{ab}{}_c\right) A_a B_b C^c = \begin{cases} \tilde{X}^{ab}{}_c\left(\phi_p^{-1} A_a\right)\left(\phi_p^{-1} B_b\right)\left(\phi_p^{-1} C^c\right); & A_a, B_b, C^c \text{ are tangent to S} \\ 0; & \text{if } A_a, B_b, \text{or } C^c \text{ is normal to S} \end{cases}$$

Again, $\phi_p\left(\tilde{X}^{ab}{}_c\right)$ is tangent to S by the second clause above.

One can also show that ϕ_p commutes with all tensor operations, i.e. addition, outer multiplication, index substitution, contraction.

For example,

$$\phi_p\left(\tilde{X}^a \tilde{Y}^b\right) = \phi_p\left(\tilde{X}^a\right) \phi_p\left(\tilde{Y}^b\right)$$

In short, at any point p in S, the tensor algebra of S-tensors is isomorphic to the tensor algebra of M-tensors that are tangent to S. With every S-tensor, I get an evil

twin M-tensor of the same index structure.

Now, this can be extended to the whole of S, and we have a map ϕ that takes any S-tensor field to an M-tensor field that is tangent to S and have the same index structure, and they also follow the same tensor algebra on both sides of the "river".

4.2.4 S-smoothness and M-smoothness

First of all, let us recall a very handy notion of the smoothness of a tensor field. Say, we have a manifold M, and an atlas $\mathcal{A}$. Now, a tensor field $T^a_{\ b}$ is smooth iff the components are smooth functions with respect to any coordinatization of the manifold with charts from the atlas $\mathcal{A}$. This means the idea of smoothess is tied to the atlas of a manifold.

Now think of the manifold S. Let me define by O an open set of M that covers S, i.e. S is a subset of O. This is going to be our working definition of O for quite some time in this discussion, unless otherwise specified. An M-tensor field T tangent to S can be smooth in two senses. If an extension of T on O, named $\overset{+}{T}$, is smooth wrt $\mathcal{A}_M$, then T is M-smooth. If, on the other hand, the S-tensor field $\phi^{-1}T$ is smooth wrt $\mathcal{A}_S$, then T is S-smooth. Now, there is a theorem that says for any M-tensor field T tangent to S, it is M-smooth iff it is S-smooth. So saying that T (when it is tangent to S) is smooth means it is smooth in both ways. On the other hand if T is not tangent to S, then it can only be M-smooth.

4.2.5 h_{ab}

This is the star of the show. We call it a projector for reasons to become clear soon. Let us define it first. Say there is a metric g_{ab} on M. We define the "induced metric" or "first fundamental form" on S by taking the pull back of g_{ab} wrt to the identity map. The induced metric, an S-field, $\tilde{h}_{ab} = id^* g_{ab}$. Then, we define the smooth symmetric M-tensor h_{ab} which is tangent to S as the ϕ-image of this induced metric $\tilde{h}_{ab}$, so that $h_{ab} = \phi\left(\tilde{h}_{ab}\right) = \phi\left(id^* g_{ab}\right)$.

This is as counterintuitive and hard to remember as possible. So we actually use an equivalent definition of h_{ab}. Here, we define h_{ab} by the way it acts on M-vectors.

$$h_{ab}X^aY^b = \begin{cases} g_{ab}X^aY^b; & \text{if } X, Y \text{ are tangent to S} \\ 0; & \text{if any of } X, Y \text{ are normal to S} \end{cases} \tag{4.1}$$

Let me show that the equivalence works. Lets say both X and Y are tangent to S. Then,

$$\begin{aligned} h_{ab}X^aY^b &= \phi\left(id^*g_{ab}\right)X^aY^b \\ &= \left(id^*g_{ab}\right)\left(\phi^{-1}X^a\right)\left(\phi^{-1}Y^b\right) \\ &= g_{ab}\ id_*\left(\phi^{-1}X^a\right)\ id_*\left(\phi^{-1}Y^b\right) \\ &= g_{ab}\ \phi\left(\phi^{-1}X^a\right)\phi\left(\phi^{-1}Y^b\right) \\ &= g_{ab}X^aY^b \end{aligned}$$

If any of X or Y is not tangent, then by virtue of h_{ab} being tangent to S, $h_{ab}X^aY^b$ is 0.

We define a perpendicular projection tensor k_{ab} too which is a smooth symmetric M-tensor field.

$$k_{ab} = g_{ab} - h_{ab} \tag{4.2}$$

Notice that k_{ab} is normal to S. If X is a M-vector tangent to S, then, $k_{ab}X^a = g_{ab}X^a - h_{ab}X^a = g_{ab}X^a - g_{ab}X^a = 0$. So k_{ab} is normal to S in the first index. But being a symmetric tensor, it must be also normal in the second index too, so it is normal as a whole.

The following propositions show that it is okay to call h_{ab} a tangential projector and k_{ab} a perpendicular or normal projector. notice that k_{ab} is normal to S.

Proposition 4.9. X^a is tangent to S $\iff$ $h^a{}_bX^b = X^a$ $\iff$ $k^a{}_bX^b = 0$

Proof. We will show the first equivalence and then the second equivalence.

First equivalence:

Let X be tangent to S.

I will show that the action of X^a and $h^a{}_bX^b$ is the same on any M-covector η_a. If η_a is normal to S, then both sides give zero. If η_a is tangent to S, then $h^a{}_bX^b\eta_a = h_{ab}\eta^aX^b = g_{ab}\eta^aX^b = X^a\eta_a$.

Conversely, let $h^a{}_bX^b = X^a$. I need to show that X^a is tangent to S. Take η_a normal to S. Then $\eta_aX^a = \eta_ah^a{}_bX^b = 0$. The last equality follows from the fact that h_{ab} is tangent in both indices.

This completes the proof for the first equivalence.

Second equivalence:

Let $h^a{}_bX^b = X^a$.

Then, $k^a{}_bX^b = \left(g^a{}_b - h^a{}_b\right)X^b = X^a - X^a = 0$

Conversely, let, $k^a{}_bX^b = 0$.

Then, $h^a{}_bX^b = \left(g^a{}_b - k^a{}_b\right)X^b = X^a - 0 = X^a$

This completes the proof of the second equivalence.

□

Proposition 4.10. X^a is normal to S $\iff$ $k^a{}_b X^b = X^a$ $\iff$ $h^a{}_b X^b = 0$

Proof. Similar as before. □

While I stated proposition 4.9 and 4.10 for a (1,0) tensor field, i.e. a vector field X^a, these propositions actually hold for tensor fields of arbitrary index structure. For example, if $T^{ab}{}_c$ is tangent to S in the second upper index, then $h^b{}_m T^{am}{}_c = T^{ab}{}_c$ and $k^b{}_m T^{am}{}_c = 0$.

Proposition 4.11. $h^a{}_b h^b{}_c = h^a{}_c$

Let me pause here for a moment, and fill up the rest of the proofs up to the Gauss Codazzi equations

4.2.6 Derivatives on S

To define a derivative operator $\tilde{D}$ on S-tensors is not hard. Since there is a metric $\tilde{h}$ on S, we can define a torsionless covariant derivative which is compatible with that metric. It is also possible to make a derivative opertor D that acts on M-tensors tangent to S and which in a way mimics the action of $\tilde{D}$ on S-tensors.

We will define D in terms of ∇. To use ∇ on a field $T^a{}_b$ defined on S, we need to know how $T^a{}_b$ changes when we move in directions away from S, which is not well defined because $T^a{}_b$ is not defined away from S. The following proposition rescues us.

Proposition 4.12. Let $\overset{1}{T}{}^a{}_b$ and $\overset{2}{T}{}^a{}_b$ be two smooth fields on an open set $O \subset M$ that agree on S. Then on any point $p \in S \cap O$,

$$h^m{}_n \nabla_m \overset{1}{T}{}^a{}_b = h^m{}_n \nabla_m \overset{2}{T}{}^a{}_b$$

The key message is that we can simply extend a field to an open set O (such that $S \subseteq O$) and then define the derivative of $T^a{}_b$ on S by $h^m{}_n \nabla_m T^a{}_b = h^m{}_n \nabla_m \overset{+}{T}{}^a{}_b$. It does not matter how we extend $T^a{}_b$ to $\overset{+}{T}{}^a{}_b$.

For tensors fields tangent to S, and defined on S, we define the derivative operator D as follows,

(1) $D_{fX+Z}Y = fD_X Y + D_Z Y$. This actually is the familiar condition (D1) for a covariant derivative.

(2) $D_a T^b{}_c = h^m{}_a h^b{}_p h^n{}_c \nabla_m T^p{}_n$. By construction, $D_a T^b{}_c$ is tangent to S.

D itself defines a legitimate derivative operator. We can prove (D1) to (D4) for the operator D.

(D1) Part of the definition.

(D2) For tensors A and B tangent to S, $D_X(aA+bB) = aD_XA + bD_XB$. This directly follows from how ∇ behaves.

$$\begin{aligned} D_s(aA^{mn}+bB^{mn}) &= h^p{}_s h^m{}_q h^n{}_r \nabla_p (aA^{qr}+bB^{qr}) \\ &= ah^p{}_s h^m{}_q h^n{}_r \nabla_p A^{qr} + bh^p{}_s h^m{}_q h^n{}_r \nabla_p B^{qr} \\ &= aD_s A^{mn} + bD_s B^{mn} \end{aligned}$$

(D3) $D_X(U\otimes V) = (D_XU)\otimes V + U\otimes D_XV$. This is a Consequence of the Leibniz rule for ∇.

$$\begin{aligned} D_m\left(U^aV^b\right) &= h^n{}_m h^a{}_p h^b{}_q \nabla_n (U^pV^q) \\ &= h^n{}_m h^a{}_p h^b{}_q (\nabla_n U^p) V^q + h^n{}_m h^a{}_p h^b{}_q U^p (\nabla_n V^q) \\ &= (D_mU^a)V^b + U^a\left(D_mV^b\right) \end{aligned}$$

(D4)$D_Xf = X^mD_mf = X^mh^n{}_m\nabla_n f = X^mh^n{}_m\nabla_n \overset{+}{f}$. Now imagine that we are at a point $p\in S$ and working in a chart where only X^1 to X^k is non-zero, i.e. $\{\partial_i\}_{(i=1,\ldots,k)}$ form a basis for $\mathscr{T}_p$. So, in an expansion like $X^m\partial_m f$ if $m>k$ then it is zero. So, the derivatives of $\overset{+}{f}$ in normal directions to S does not matter, and we can do good enough with only $\partial_n f$ with $n=1,\ldots,k$ as these are well defined even without the extension of f. So, $D_Xf = X^mh^n{}_m\partial_n\overset{+}{f} = X^n\partial_n f = X(f)$. Since we are dealing with a scalar, it is true regardless of which chart I am using and the result is true chart independently.

Another property of D is quite important. D is metric compatible in the following sense,

Proposition 4.13. $D_Xh = 0$ or $D_ah_{bc} = 0$

Proof.

$$\begin{aligned} D_ah_{bc} &= h^m{}_a h^n{}_b h^p{}_c \nabla_m h_{np} \\ &= h^m{}_a h^n{}_b \left[\nabla_m\left(h^p{}_c h_{np}\right) - h_{np}\nabla_m h^p{}_c\right] \\ &= h^m{}_a h^n{}_b \left(\nabla_m h_{cn} - h_{np}\nabla_m h^p{}_c\right) \\ &= h^m{}_a h^n{}_b \left(\nabla_m h_{cn} - h^p{}_n\nabla_m h_{cp}\right) \\ &= h^m{}_a h^n{}_b \nabla_m h_{cn} - h^m{}_a h^p{}_b \nabla_m h_{pc} \\ &= h^m{}_a h^n{}_b \nabla_m h_{nc} - h^m{}_a h^n{}_b \nabla_m h_{nc} \\ &= 0 \end{aligned}$$

□

$\overset{D}{T}$ is a $(1,2)$-tensor defined as follows:- $\overset{D}{T}(\omega, X, Y) = 0$ if any of ω, X, Y are normal to S. On vectors and covectors tangent to S, $\overset{D}{T}$ acts as,

$$\overset{D}{T}(\omega, X, Y) = \omega\left(D_X Y - D_Y X - [X, Y]\right) \tag{4.3}$$

Proposition 4.14. $\overset{D}{T}$ is identically 0 on S.

Proof. First of all, $\overset{D}{T}(\omega, X, Y) = \omega\left(D_X Y - D_Y X - [X, Y]\right)$. Now, we can extend X, Y to $\overset{+}{X}, \overset{+}{Y}$. That extension will still coincide with X, Y on S. We also have $[X, Y] = [\overset{+}{X}, \overset{+}{Y}]$ on S.

Based on this,

$$\begin{aligned}
\overset{D}{T}(\omega, X, Y) &= \omega\left(D_X Y - D_Y X - [X, Y]\right) \\
&= \omega_a\left((D_X Y)^a - (D_Y X)^a - [X, Y]^a\right) \\
&= \omega_a\left(X^m D_m Y^a - Y^m D_m X^a - [X, Y]^a\right) \\
&= \omega_a\left(\overset{+}{X}{}^m h^n{}_m h^a{}_p \nabla_n \overset{+}{Y}{}^p - \overset{+}{Y}{}^m h^n{}_m h^a{}_p \nabla_n \overset{+}{X}{}^p - [\overset{+}{X}, \overset{+}{Y}]^a\right) \\
&= \omega_p \overset{+}{X}{}^n \nabla_n \overset{+}{Y}{}^p - \omega_p \overset{+}{Y}{}^n \nabla_n \overset{+}{X}{}^p - \omega_p [\overset{+}{X}, \overset{+}{Y}]^p \\
&= \omega_p\left(\nabla_{\overset{+}{X}} \overset{+}{Y} - \nabla_{\overset{+}{Y}} \overset{+}{X} - [\overset{+}{X}, \overset{+}{Y}]\right)^p \\
&= \omega\left(\nabla_{\overset{+}{X}} \overset{+}{Y} - \nabla_{\overset{+}{Y}} \overset{+}{X} - [\overset{+}{X}, \overset{+}{Y}]\right) \\
&= T\left(\omega, \overset{+}{X}, \overset{+}{Y}\right) \quad \text{(where } T \text{ is regular torsion)} \qquad = 0
\end{aligned}$$

□

Now, we have defined both the unique torsionless metric compatible derivative operator $\tilde{D}$ on S-tensors and the operator D on M-tensors on S that are tangent to S. The basic result that we will now show is the following,

Proposition 4.15. $\phi\left(\tilde{D}_{\tilde{X}} \tilde{T}\right) = D_X T$ where $X = \phi\left(\tilde{X}\right)$ and $T = \phi\left(\tilde{T}\right)$

Proof. We define $\mathcal{D}_{\tilde{X}} \tilde{T}$ as follows,

$$\mathcal{D}_{\tilde{X}} \tilde{T} = \phi^{-1}\left(D_X T\right)$$

where $X = \phi(\tilde{X})$ and $T = \phi(\tilde{T})$. Our way to prove the proposition is to show that $\mathcal{D}$ is a torsionless metric compatible derivative operator and by virtue of the uniqueness of such operators $\mathcal{D} = \tilde{D}$. To show that $\mathcal{D}$ is a covariant derivative on S, we need

to show that it follow (D1) to (D4). The key underlying fact here is that ϕ is an isomorphism and it preserves the structure of tensor algebra.

(D1)

$$\begin{aligned}\mathcal{D}_{f\tilde{X}+\tilde{Y}}\tilde{Z} &= \phi^{-1}\left(D_{fX+Y}Z\right)\\ &= \phi^{-1}\left(fD_XZ + D_YZ\right)\\ &= f\phi^{-1}\left(D_XZ\right) + \phi^{-1}\left(D_YZ\right)\\ &= f\mathcal{D}_{\tilde{X}}\tilde{Z} + \mathcal{D}_{\tilde{Y}}\tilde{Z}\end{aligned}$$

(D2)

$$\mathcal{D}_{\tilde{X}}\left(a\tilde{A} + b\tilde{B}\right) = a\mathcal{D}_{\tilde{X}}\tilde{A} + b\mathcal{D}_{\tilde{X}}\tilde{B}$$

(D3)

$$\begin{aligned}\mathcal{D}_{\tilde{X}}\left(\tilde{A}\otimes\tilde{B}\right) &= \phi^{-1}\left(D_X\left(A\otimes B\right)\right)\\ &= \phi^{-1}\left(D_XA\otimes B + A\otimes D_XB\right)\\ &= \phi^{-1}\left(D_XA\right)\otimes\tilde{B} + \tilde{A}\otimes\phi^{-1}\left(D_XB\right)\\ &= \mathcal{D}_{\tilde{X}}\tilde{A}\otimes\tilde{B} + \tilde{A}\otimes\mathcal{D}_{\tilde{X}}\tilde{B}\end{aligned}$$

(D4)

$$\begin{aligned}\mathcal{D}_{\tilde{X}}f &= \phi^{-1}\left(D_Xf\right)\\ &= \phi^{-1}\left(X(f)\right)\\ &= X(f) \quad \text{(since } X(f) \text{ is a scalar function)} \qquad\qquad = \tilde{X}(f)\end{aligned}$$

Thinking about the meaning of a vector and the fact that both $\tilde{X}$ and X are tangent vectors to the same curve will convince us of the last equality

More over, $\mathcal{D}$ follows the torsionless condition,

(DT)

$$\begin{aligned}\tilde{T}\left(\tilde{\omega},\tilde{X},\tilde{Y}\right) &= \tilde{\omega}\left(\mathcal{D}_{\tilde{X}}\tilde{Y} - \mathcal{D}_{\tilde{Y}}\tilde{X} - [\tilde{X},\tilde{Y}]\right)\\ &= \omega\left(\phi\left(\mathcal{D}_{\tilde{X}}\tilde{Y} - \mathcal{D}_{\tilde{Y}}\tilde{X} - [\tilde{X},\tilde{Y}]\right)\right)\\ &= \omega\left(D_XY - D_YX - [X,Y]\right)\\ &= \overset{D}{T}(\omega,X,Y)\\ &= 0\end{aligned}$$

And $\mathcal{D}$ is metric compatible.

(DC)

$$\begin{aligned}\mathcal{D}_{\tilde{X}}\tilde{h} &= \phi^{-1}\left(D_X h\right)\\ &= \phi^{-1}(0)\\ &= 0\end{aligned}$$

So, $\mathcal{D}$ is the unique metric compatible torsionless operator on S so we denote it by $\tilde{D}$, and we have the result,

$$\phi\left(\tilde{D}_{\tilde{X}}\tilde{T}\right) = D_X T$$

□

4.2.7 Extrinsic Curvature

Now that we are sufficiently familiar with the notion of derivatives on a submanifold, we can turn to the concept of extrinsic curvature. Intuitively, extrinsic curvature should tell us how bend a surface is on an ambient space. Before elaborating on how exactly the extrinsic curvature tells us about the bending of a submanifold, let us define the quantity mathematically first.

We know that,

$$\begin{aligned}h^m{}_a h^n{}_b h^p{}_c \nabla_m h_{np} &= 0\\ h^m{}_a k^n{}_b k^p{}_c \nabla_m h_{np} &= 0\end{aligned}$$

However, the hhk projection, $h^m{}_a h^n{}_b k^p{}_c \nabla_m h_{np}$ is not zero and we define a new quantity, called the extrinsic curvature of S, as follows,

$$K_{abc} = h^m{}_a h^n{}_b k^p{}_c \nabla_m h_{np}$$

K_{abc} has particular geometric significance, and we will show some of those significance in this section.

4.2.8 Gauss Codazzi Equations

Before proving those equations, let me remind one useful way of characterizing the Riemann tensor. If v^a is a vector field, then,

$$
\begin{aligned}
[\nabla_c\,,\nabla_d]v^a &= R^a{}_{mcd}v^m \\
2\nabla_{[c}\nabla_{d]}v^a &= R^a{}_{mcd}v^m \\
\nabla_{[c}\nabla_{d]}v^a &= \frac{1}{2}R^a{}_{mcd}v^m
\end{aligned}
$$

This can also be generalized, in the following way. Say, $T^a{}_b$ is a (1,1) field. Then the antisymmetrized double covariant derivative on it can be expanded in temrs of the Riemann tensor as follows,

$$
2\nabla_{[c}\nabla_{d]}T^a{}_b = R^a{}_{mcd}T^m{}_b - R^m{}_{bcd}T^a{}_m
$$

One final fact about the Riemann tensors that needs a reminder is, $R_{abcd} = R_{cdab}$.

We will use a script R, $\mathcal{R}$, to denote the Riemann tensor defined wrt to D. This is going to be a tangent tensor field. So, we define the Riemann tensor to be, $\mathcal{R}^a{}_{mcd}v^m = 2D_{[c}D_{d]}v^a$ where v^a is an M-field tangent to S.

To Do: Write down some more words about the proper definition of $\mathcal{R}$

Now,

$$
\begin{aligned}
D_{[c}D_{d]}v^a &= h^m{}_{[c}h^n{}_{d]}h^a{}_p\nabla_m D_n v^p \\
&= h^m{}_{[c}h^n{}_{d]}h^a{}_p\nabla_m\left(h^q{}_n h^p{}_r\nabla_q v^r\right) \\
&= h^m{}_{[c}h^n{}_{d]}h^a{}_p\left(\nabla_m h^q{}_n\right)h^p{}_r\nabla_q v^r + h^m{}_{[c}h^n{}_{d]}h^a{}_p h^q{}_n\left(\nabla_m h^p{}_r\right)\nabla_q v^r \\
&+ h^m{}_{[c}h^n{}_{d]}h^a{}_p h^q{}_n h^p{}_r\nabla_m\nabla_q v^r \\
&= h^m{}_{[c}h^n{}_{d]}h^a{}_r\left(\nabla_m h^q{}_n\right)\nabla_q v^r + h^m{}_{[c}h^q{}_{d]}h^a{}_p\left(\nabla_m h^p{}_r\right)\nabla_q v^r \\
&+ h^m{}_{[c}h^q{}_{d]}h^a{}_r\nabla_m\nabla_q v^r \\
&\left(\text{ Mostly using } h^a{}_b h^b{}_c = h^a{}_c\right)
\end{aligned}
\tag{4.4}
$$

Now, let's look at them term by term.

First Term:

$$
\begin{aligned}
&h^m{}_{[c}h^n{}_{d]}h^a{}_r\left(\nabla_m h^q{}_n\right)\nabla_q v^r \\
&= h^m{}_{[c}h^n{}_{d]}h^a{}_r\left(\nabla_m h_{nq}\right)\nabla^q v^r \\
&= h^a{}_r h^m{}_{[c}h^n{}_{d]}\left(\nabla_m h_{nq}\right)\nabla^q v^r \\
&= -h^a{}_r K_{[cd]q}\nabla^q v^r
\end{aligned}
$$

(Using the relation $K_{abc} = -h^m{}_a h^n{}_b\nabla_m h_{nc}$)

$= 0$

(Since the antisymmetrization of the first two indices of K_{abc} gives zero.)

Second Term:

$$
\begin{aligned}
& h^m{}_{[c} h^q{}_{d]} h^a{}_p \left(\nabla_m h^p{}_r\right) \nabla_q v^r \\
&= h^m{}_{[c} h^q{}_{d]} h^{ap} \left(\nabla_m h_{pr}\right) \nabla_q v^r \\
&= h^m{}_{[c} h^q{}_{d]} h^{pa} \left(\nabla_m h_{pr}\right) \nabla_q v^r \\
&= g^{sa} h^m{}_{[c} h^q{}_{d]} h^p{}_s \left(\nabla_m h_{pr}\right) \nabla_q v^r \\
&= g^{sa} h^m{}_{[c} h^p{}_{|s|} h^q{}_{d]} \left(\nabla_m h_{pr}\right) \nabla_q v^r \\
&= -g^{sa} K_{[c|sr|} h^q{}_{d]} \nabla_q v^r \\
&= -g^{sa} K_{[c|sr|} h^q{}_{d]} \nabla_q \left(h^r{}_t v^t\right) \\
&\text{(The trick is to realize that v is tangent to S.)} \\
&= -K_{[c}{}^a{}_{|r|} h^q{}_{d]} \nabla_q \left(h^r{}_t v^t\right) \\
&= -K_{[c}{}^a{}_{|r|} h^q{}_{d]} \left(\nabla_q h^r{}_t\right) v^t - K_{[c}{}^a{}_{|r|} h^q{}_{d]} h^r{}_t \nabla_q v^t \\
&= -K_{[c}{}^a{}_{|r|} h^q{}_{d]} \left(\nabla_q h^r{}_t\right) v^t - 0 \\
&\left(\begin{array}{l}\text{The second term goes to zero as the extrinsic curvature } K_{abc} \text{ is} \\ \text{normal in the third index.}\end{array}\right) \\
&= -K_{[c}{}^a{}_{|r|} h^q{}_{d]} \left(\nabla_q h^r{}_t\right) \left(h^t{}_u v^u\right) \\
&= -K_{[c}{}^{ar} h^q{}_{d]} \left(\nabla_q h_{tr}\right) \left(h^t{}_u v^u\right) \\
&= -K_{[c}{}^{ar} h^q{}_{d]} h^t{}_u \left(\nabla_q h_{tr}\right) v^u \\
&= K_{[c}{}^{ar} K_{d]ur} v^u
\end{aligned}
$$

Third Term:

$$
\begin{aligned}
& h^m{}_{[c} h^q{}_{d]} h^a{}_r \nabla_m \nabla_q v^r \\
&= h^m{}_c h^q{}_d h^a{}_r \nabla_{[m} \nabla_{q]} v^r \\
&= h^m{}_c h^q{}_d h^a{}_r \left(\frac{1}{2}\right) R^r{}_{nmq} v^n \\
&= \frac{1}{2} R^r{}_{nmq} h^m{}_c h^q{}_d h^a{}_r v^n
\end{aligned}
$$

Collecting the terms, I get from eq. (4.4),

$$D_{[c}D_{d]}v^a = \frac{1}{2}R^r{}_{nmq}h^m{}_c h^q{}_d h^a{}_r v^n + K_{[c}{}^{ar}K_{d]ur}v^u$$
$$= \frac{1}{2}R^r{}_{mnq}h^n{}_c h^q{}_d h^a{}_r v^m + K_{[c}{}^{ar}K_{d]mr}v^m$$

Now, from our earlier characterization of $\mathcal{R}$, we have,

$$\mathcal{R}^a{}_{mcd}v^m = 2D_{[c}D_{d]}v^a$$
$$\implies \mathcal{R}^a{}_{mcd}v^m = R^r{}_{mnq}h^n{}_c h^q{}_d h^a{}_r v^m + 2K_{[c}{}^{ar}K_{d]mr}v^m$$

Now, taking any vector η, not necessarily tangent, and then taking the tangential projection of that vector, $v^m = h^m{}_s\eta^s$,

$$\mathcal{R}^a{}_{mcd}h^m{}_s\eta^s = R^r{}_{mnq}h^n{}_c h^q{}_d h^a{}_r h^m{}_s\eta^s + 2K_{[c}{}^{ar}K_{d]mr}h^m{}_s\eta^s$$
$$\mathcal{R}^a{}_{mcd}h^m{}_s\eta^s = \left(R^r{}_{mnq}h^n{}_c h^q{}_d h^a{}_r h^m{}_s + 2K_{[c}{}^{ar}K_{d]mr}h^m{}_s\right)\eta^s$$
$$\mathcal{R}^a{}_{mcd}h^m{}_s = R^r{}_{mnq}h^n{}_c h^q{}_d h^a{}_r h^m{}_s + 2K_{[c}{}^{ar}K_{d]mr}h^m{}_s$$
$$\mathcal{R}^a{}_{scd} = R^r{}_{mnq}h^n{}_c h^q{}_d h^a{}_r h^m{}_s + 2K_{[c}{}^{ar}K_{d]sr}$$

(We assumed that $\mathcal{R}$ is a tangent field, also K is tangent in the second index.)

$$\mathcal{R}^a{}_{bcd} = R^r{}_{mnq}h^n{}_c h^q{}_d h^a{}_r h^m{}_b + 2K_{[c}{}^{ar}K_{d]br}$$

$(s \to b)$

$$\mathcal{R}^a{}_{bcd} = R^m{}_{rnq}h^n{}_c h^q{}_d h^a{}_m h^r{}_b + 2K_{[c}{}^{am}K_{d]bm}$$

$(r \leftrightarrow m)$

$$\mathcal{R}^a{}_{bcd} = R^m{}_{npq}h^p{}_c h^q{}_d h^a{}_m h^n{}_b + 2K_{[c}{}^{am}K_{d]bm}$$

$(r \to n, n \to p)$

$$\mathcal{R}^a{}_{bcd} = R^m{}_{npq}h^a{}_m h^n{}_b h^p{}_c h^q{}_d + 2K_{[c}{}^{am}K_{d]bm}$$

And let me put this in a box, as I have to memorize this as the Make the box. Then start deriving the second Gauss Codazzi relation.

The second Gauss Codazzi equation is

$$h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r \nabla_m K_{abc} = \frac{1}{2}h^m{}_n h^a{}_p h^b{}_q h^c{}_r R_{mabc} \tag{4.5}$$

To prove it, let's start with the LHS,

$$LHS = h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r \nabla_m K_{abc}$$

$$
\begin{aligned}
&= h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r \nabla_m \left(-h^s{}_a h^t{}_b \nabla_s h_{tc}\right) \\
&= -h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r \left(\nabla_m h^s{}_a\right) h^t{}_b \nabla_s h_{tc} - h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r h^s{}_a \left(\nabla_m h^t{}_b\right) \nabla_s h_{tc} \\
&- h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r h^s{}_a h^t{}_b \nabla_m \nabla_s h_{tc} \\
&\equiv -A - B - C \text{ (Naming the three terms.)}
\end{aligned}
$$

Term A:

$$
\begin{aligned}
A &= h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r \left(\nabla_m h^s{}_a\right) h^t{}_b \nabla_s h_{tc} \\
&= h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r \left(\nabla_m h_{as}\right) h^t{}_b \nabla^s h_{tc} \\
&= h^m{}_{[n}h^a{}_{p]}h^t{}_q k^c{}_r \left(\nabla_m h_{as}\right) \left(\nabla^s h_{tc}\right) \\
&= -K_{[np]s}h^t{}_q k^c{}_r \nabla^s h_{tc} \\
&= 0
\end{aligned}
$$

Term B:

$$
\begin{aligned}
B &= h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r h^s{}_a \left(\nabla_m h^t{}_b\right) \nabla_s h_{tc} \\
&= h^m{}_{[n}h^s{}_{p]}h^b{}_q k^c{}_r \left(\nabla_m h^t{}_b\right) \nabla_s h_{tc} \\
&= h^m{}_{[n}h^s{}_{p]}h^b{}_q k^c{}_r \left(\nabla_m h_{bt}\right) \nabla_s h^t{}_c \\
&= -K_{[n|qt|}h^s{}_{p]}k^c{}_r \nabla_s h^t{}_c \\
&= -k^u{}_t K_{[n|qu|}h^s{}_{p]}k^c{}_r \nabla_s h^t{}_c \\
&\text{(K is normal in the third index, so introducing k makes sense.)} \\
&= -k^{tu} K_{[n|qu|}h^s{}_{p]}k^c{}_r \nabla_s h_{tc} \\
&= -k^t{}_u K_{[n}{}^u{}_{|q|}h^s{}_{p]}k^c{}_r \nabla_s h_{tc} \\
&= -K_{[n}{}^u{}_{|q|}h^s{}_{p]}k^t{}_u k^c{}_r \nabla_s h_{tc} \\
&= 0
\end{aligned}
$$

Term C:

$$
\begin{aligned}
C &= h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r h^s{}_a h^t{}_b \nabla_m \nabla_s h_{tc} \\
&= h^m{}_{[n}h^s{}_{p]} \left(\nabla_m \nabla_s h_{tc}\right) h^b{}_q h^t{}_b k^c{}_r \\
&= h^m{}_n h^s{}_p \left(\nabla_{[m}\nabla_{s]}h_{tc}\right) h^b{}_q h^t{}_b k^c{}_r \\
&= h^m{}_n h^s{}_p h^b{}_q h^t{}_b k^c{}_r \left(\nabla_{[m}\nabla_{s]}h_{tc}\right) \\
&= \frac{1}{2} h^m{}_n h^s{}_p h^b{}_q h^t{}_b k^c{}_r \left(-R^u{}_{tms}h_{uc} - R^u{}_{cms}h_{tu}\right)
\end{aligned}
$$

$$
\begin{aligned}
&= -\frac{1}{2}\left(h^m{}_n h^s{}_p h^b{}_q h^t{}_b \left(k^c{}_r h^u{}_c\right) R_{utms} + h^m{}_n h^s{}_p h^b{}_q h^t{}_b k^c{}_r h^u{}_t R_{ucms}\right) \\
&= -\frac{1}{2}\left(0 + h^m{}_n h^s{}_p h^u{}_q k^c{}_r R_{ucms}\right) \\
&= -\frac{1}{2} h^m{}_n h^s{}_p h^u{}_q k^c{}_r R_{msuc} \\
&= -\frac{1}{2} h^m{}_n h^a{}_p h^b{}_q k^c{}_r R_{mabc}
\end{aligned}
$$

In short,

$$
h^m{}_{[n} h^a{}_{p]} h^b{}_q k^c{}_r \nabla_m K_{abc} = \frac{1}{2} h^m{}_n h^a{}_p h^b{}_q k^c{}_r R_{mabc}
$$

Make another box for this equation.

4.3 Hypersurfaces

$$
h_{ab} = g_{ab} - \varepsilon n_a n_b \tag{4.6}
$$

$$
k_{ab} = \varepsilon n_a n_b \tag{4.7}
$$

This is shown by testing this definition of $h_{ab} = g_{ab} - \varepsilon n_a n_b$ against the main definition.

Take a vector $X \in T$. I expect $h_{ab} X^b$ to be $g_{ab} X^b$. Now,

$$
\begin{aligned}
h_{ab} X^b &= \left(g_{ab} - \varepsilon n_a n_b\right) X^b \\
&= g_{ab} X^b - \varepsilon n_a \underbrace{n_b X^b}_{=0} \\
&\quad (X \text{ is tangent to S, } n \text{ is normal}) \\
&= g_{ab} X^b
\end{aligned}
$$

For a vector $Y \in N$, I expect $h_{ab} Y^b = 0$. It suffices to simply take $n^a \in N$ as N is 1 dimensional anyways.

$$
\begin{aligned}
h_{ab} n^b &= \left(g_{ab} - \varepsilon n_a n_b\right) n^b \\
&= g_{ab} n^b - \varepsilon n_a \left(n_b n^b\right) \\
&= n_a - \varepsilon^2 n_a \\
&= n_a - n_a \qquad\qquad = 0
\end{aligned}
$$

Getting the form of k_{ab} is easy.

$$k_{ab} = g_{ab} - h_{ab} = g_{ab} - (g_{ab} - \varepsilon n_a n_b) = \varepsilon n_a n_b$$

Define,

$$K_{ab} = K_{abc} n^c \tag{4.8}$$

In the context of hypersurfaces, this is often called the "Extrinsic Curvature" or "2nd Fundamental Form".

Another way K_{ab} is expressed,

$$K_{ab} = -h^p{}_a h^q{}_b \nabla_p n_q \tag{4.9}$$

$$\begin{aligned}
K_{ab} &= K_{abc} n^c \\
&= h^p{}_a h^q{}_b \left(\nabla_p h_{qc} \right) n^c \\
&= h^p{}_a h^q{}_b \left(\nabla_p \left(\underbrace{h_{qc} n^c}_{=0} \right) - h_{qc} \nabla_p n^c \right) \\
&= -h^p{}_a h^q{}_b h_{qc} \nabla_p n^c \\
&= -h^p{}_a h^c{}_b \nabla_p n_c \\
&= -h^p{}_a h^q{}_b \nabla_p n_q
\end{aligned}$$

$K_{[ab]} = 0$; K_{ab} is symmetric.

$$K_{[ab]} = K_{[ab]c} n^c = 0 \tag{4.10}$$

$$K_{abc} = \varepsilon K_{ab} n_c \tag{4.11}$$

We show it by acting on both sides with an arbitrary vector. In this case, it suffices to act with $X^a \in T$ and n^a.

First, act with $X^a \in T$.

$$\begin{aligned} \text{LHS} &\implies K_{abc}X^c = 0 \\ &\quad (K_{abc} \text{ is normal in the index c}) \\ \text{RHS} &\implies \varepsilon K_{ab} n_c X^c = 0 \end{aligned}$$

Then, act with n^a,

$$\begin{aligned} \text{LHS} &\implies K_{abc}n^c = K_{ab} \\ \text{RHS} &\implies \varepsilon K_{ab} n_c n^c = \varepsilon K_{ab}\varepsilon = \varepsilon^2 K_{ab} = K_{ab} \end{aligned}$$

Yet another way to express K_{ab} is,

$$K_{ab} = -\frac{1}{2}\mathcal{L}_n h_{ab} \tag{4.12}$$

4.3.1 Gauss-Codazzi eqns for hypersurface

The first GC eqn was,

$$\begin{aligned} \mathcal{R}^a{}_{bcd} &= R^m{}_{npq} h^a{}_m h^n{}_b h^p{}_c h^q{}_d + 2K_{[c}{}^{am} K_{d]bm} \\ \implies \mathcal{R}_{abcd} &= R_{mnpq} h^m{}_a h^n{}_b h^p{}_c h^q{}_d + 2K_{[c|a|}{}^m K_{d]bm} \end{aligned} \tag{4.13}$$

Now, I use the eqn $K_{abc} = \varepsilon K_{ab} n_c$ in eq. (4.13). Start with the second term.

$$\begin{aligned} K_{[c|a|}{}^m K_{d]bm} &= \varepsilon K_{[c|a|} n^m \, \varepsilon K_{d]b} n_m \\ &= n^m n_m \varepsilon^2 K_{[c|a|} K_{d]b} \\ &= -\varepsilon K_{a[c} K_{d]b} \end{aligned}$$

So, eq. (4.13) $\implies$

$$\mathcal{R}_{abcd} = R_{mnpq} h^m{}_a h^n{}_b h^p{}_c h^q{}_d - 2\varepsilon K_{a[c} K_{d]b} \tag{4.14}$$

The second GC eqn was,

$$h^m{}_{[p}h^a{}_{q]}h^b{}_r k^c{}_s \nabla_m K_{abc} = -\frac{1}{2}h^m{}_p h^a{}_q h^b{}_r k^c{}_s R_{mabc} \tag{4.15}$$

Now, to substitute $K_{abc} = \varepsilon K_{ab} n_c$ in the LHS, I have,

$$\begin{aligned} &h^m{}_{[p}h^a{}_{q]}h^b{}_r k^c{}_s \nabla_m \left(\varepsilon K_{ab} n_c\right) \\ &= \varepsilon h^m{}_{[p}h^a{}_{q]}h^b{}_r k^c{}_s \left(\left(\nabla_m K_{ab}\right) n_c + K_{ab}\left(\nabla_m n_c\right)\right) \end{aligned} \tag{4.16}$$

The second term in eq. (4.16) actually goes to zero. Notice that, $n^a \nabla_b n_a = 0$. This is because $n^a n_a = \varepsilon$

$$\nabla_b \left(n^a n_a\right) = 0 \implies n^a \nabla_b n_a = 0$$

So,

$$k^c{}_s \nabla_m n_c = \varepsilon n^c n_s \nabla_m n_c = \varepsilon n_s n^c \nabla_m n_c = 0$$

And this makes the 2nd term of eq. (4.16) zero.
So, eq. (4.16) $\implies$

$$\begin{aligned} &h^m{}_{[p}h^a{}_{q]}h^b{}_r k^c{}_s \nabla_m \left(\varepsilon K_{ab} n_c\right) \\ &= \varepsilon h^m{}_{[p}h^a{}_{q]}h^b{}_r k^c{}_s n_c \nabla_m K_{ab} \\ &= \varepsilon h^m{}_{[p}h^a{}_{q]}h^b{}_r n_s \nabla_m K_{ab} \end{aligned} \tag{4.17}$$

Using this in the LHS of eq. (4.15) $\implies$

4.3.2 Integration on Hypersurfaces

Surface Element (non-Null Case)

If Σ is non-null, then we define a surface element on Σ by

$$d\Sigma = |h|^{1/2} d^3 y \tag{4.18}$$

where $h = \det\left[h_{\alpha\beta}\right]$. $n_a d\Sigma$ is a directed surface element which points in the direction of increasing Φ. In the null case neither $d\Sigma$ not $n_a d\Sigma$ is defined as $h = 0$ and n_a

does not exist. So, for the null case we need to make a generaliztion of eq. (4.18) so that it works for null hypersurfaces too. The general formula for a directed surface element is,

$$d\Sigma_a = \varepsilon_{abcd} e_1^b e_2^c e_3^d d^3y \tag{4.19}$$

where $\varepsilon_{abcd} = \sqrt{-g}[abcd]$ is the Levi-Civita tensor.

We will later show that when Σ is not null, $d\Sigma_a = \varepsilon n_a d\Sigma$. This means, for a spacelike Σ, $d\Sigma_a$ points in the direction of decreasing Φ. For timelike Σ, $d\Sigma_a$ points in the direction of increasing Φ. Before proving that $d\Sigma_a = \varepsilon n_a d\Sigma$, we need to fix a convention for $d\Sigma_a$ itself as there is an ambiguity of how we order the coordinates intrinsic to the hypersurface.

The ordering we pick is such that the scalar $f = \varepsilon_{abcd} n^a e_1^b e_2^c e_3^d$ is positive. (We will see later that $f = |h|^{1/2}$ in the non null case.) To see how this is ordering is done in practice, we give two examples in Minkowski space. First, consider a constant t hypersurface. Then, $\Phi = t = c$ (for some constant c). Then, $n_a = \frac{-\partial_a \Phi}{\sqrt{|\partial_m \Phi \partial^m \Phi|}} = \frac{-\partial_a t}{\sqrt{|\partial_m t \partial^m t|}} = -\partial_a t$. If we take the ordering $y^\alpha = (x, y, z)$, then, $f = \varepsilon_{mabc} n^m e_1^a e_2^b e_3^c = \varepsilon_{mabc}(-\partial^m t)\delta_x^a \delta_y^b \delta_z^c = \varepsilon_{txyz} = 1 > 0$

For the second example, take a constant x surface, $\Phi = x = c$. So, $n_a = \frac{\partial_a x}{\sqrt{\partial_a x \partial^a x}} = \partial_a x$. Here, the ordering $y^\alpha = (y, t, z)$ works. $f = \varepsilon_{mabc} n^m e_1^a e_2^b e_3^c = \varepsilon_{mabc}(\partial^m x)\delta_y^a \delta_t^b \delta_z^c = \varepsilon_{xytz} = -\varepsilon_{xtyz} = \varepsilon_{txyz} = 1 > 0$. Note that the more intuitive ordering $y^\alpha = (t, y, z)$ does not work. $f = \varepsilon_{mabc}(\partial^m x)\delta_t^a \delta_y^b \delta_z^c = \varepsilon_{xtyz} = -\varepsilon_{txyz} = -1 < 0$.

Next, we turn to prove that for non null hypersurface Σ, the directed surface element as defined in eq. (4.19) implies that $d\Sigma_a = \varepsilon n_a d\Sigma$. To prove this, first notice that $\varepsilon_{abcd} e_1^b e_2^c e_3^d$ must be normal to Σ (i.e. proportional to n_a) since $\varepsilon_{abcd} e_1^b e_2^c e_3^d$ pulled back to Σ gives zero. This is because the pull back of $\varepsilon_{abcd} e_1^b e_2^c e_3^d$ is $e_\alpha^a \varepsilon_{abcd} e_1^b e_2^c e_3^d$ and it is zero by virtue of the antisymmetry in ε_{abcd}. Being proportional to n_a means $\varepsilon_{abcd} e_1^b e_2^c e_3^d = \varepsilon f n_a$ where f is given by $f = \varepsilon_{abcd} n^a e_1^b e_2^c e_3^d$.

We can calculate f in any coordinate system we want. We pick the system where $x^0 = \Phi$ and on Σ (where $x^0 = 0$ without loss of generality) we identify $x^\alpha = y^\alpha$. Then $n_a \overset{*}{=} \frac{\varepsilon \partial_a x^0}{\sqrt{|g^{mn}\partial_m x^0 \partial_n x^0|}} = \left(\frac{\varepsilon}{|g^{00}|^{1/2}}, 0, 0, 0\right)$. And, $n^a \overset{*}{=} g^{ab} n_b = g^{a0} n_0 = (g^{00} n_0, 0, 0, 0) = \left(g^{00}\frac{\varepsilon}{|g^{00}|^{1/2}}, 0, 0, 0\right)$.

Also, in this case, $g(\partial_0, \partial_0) = g_{00} = \varepsilon |g_{00}|$ and $g^{00} = g^{0a} g^{0b} g_{ab} = g^{00} g^{00} g_{00} = (g^{00})^2 g_{00} \Rightarrow g^{00} = \frac{1}{g_{00}} \Rightarrow |g^{00}| = \frac{1}{|g_{00}|}$. Therefore, $g^{00} = \frac{1}{g_{00}} = \frac{1}{\varepsilon |g_{00}|} = \frac{\varepsilon}{|g_{00}|} = \varepsilon |g^{00}|$. Using this, $n^a \overset{*}{=} \left(|g^{00}|^{1/2}, 0, 0, 0\right)$. Hence, $f \overset{*}{=} \varepsilon_{mabc} n^m \delta_1^a \delta_2^b \delta_3^c = \varepsilon_{0123} |g^{00}|^{1/2} = \sqrt{|g|}[0123]|g^{00}|^{1/2} = \sqrt{|g|}|g^{00}|^{1/2}$.

Now, reminding ourselves some high-school matrix stuff, the inverse matrix A^{-1} of a matrix A is $A^{-1} = \frac{[\text{cofactor}(A)]^T}{\det(A)}$. So, $g^{00} = \frac{\text{cofactor}(g_{00})}{g}$ where cofactor(g_{00}) is the determinant of g_{ab} with the row and column of g_{00} deleted. We also notice that the intrinsic metric is given by $\tilde{h}_{\alpha\beta} = e^a_\alpha e^b_\beta g_{ab} \overset{*}{=} g_{\alpha\beta}$. So the form of the metric g_{ab} looks like,

$$g_{ab} = \begin{bmatrix} g_{00} & 0 & 0 & 0 \\ 0 & & & \\ 0 & & \tilde{h}_{\alpha\beta} & \\ 0 & & & \end{bmatrix}$$

Naturally, cofactor$(g_{00}) = \tilde{h}$. So, $f \overset{*}{=} \sqrt{|g|}|g^{00}|^{1/2} = \sqrt{|g|}|\frac{\text{cofactor}(g_{00})}{g}|^{1/2} = \sqrt{|g|}|\frac{\tilde{h}}{g}|^{1/2} = |\tilde{h}|^{1/2}$. While we proved it in a specific coordinate system, $f = |\tilde{h}|^{1/2}$ is true in any coordinate as $|\tilde{h}|$ is a scalar just like $\tilde{h}_{\alpha\beta}$ are scalars with respect to coordinate transformations of M.

To complete the proof,

$$\begin{aligned} d\Sigma_a &= \varepsilon_{abcd} e^b_1 e^c_2 e^d_3 \, d^3y \\ &= \varepsilon f n_a \, d^3y \\ &= \varepsilon |\tilde{h}|^{1/2} n_a d^3y \\ &= \varepsilon n_a d\Sigma \end{aligned}$$

Chapter 5

Gravitational Waves

I will be following several GR textbooks, often to the letter(not quite).

There are some parts of GW wave theory that I am still a bit hazy on, and I will need more clarification on them. Keeping this aspect aside, let me dive in.

5.1 The Harmonic Gauge

In GR, gauge is simply a choice of a class of coordinate system that follows a specific condition. The condition is called the “gauge condition" or “gauge choice". The harmonic gauge is defined as follows,

$$\Gamma^m \equiv g^{ab}\Gamma^m_{ab} = 0$$

Two questions immediately follow this - is it always possible to impose this gauge, and why is the name "harmonic"? I will address the first one below. As for the second one, I am not sure.

To answer the first question in affirmative, I need to show that given any coordinate system, it is always possible to go to a new coordinate where the gauge condition is satisfied. This is what is meant when it is claimed that it is always possible to impose this or that gauge condition.

We know how Γ^a_{bc} transforms under a coordinate transformation.

$$\begin{aligned}
\Gamma'^a_{bc} &= \frac{\partial x'^a}{\partial x^m}\frac{\partial x^n}{\partial x'^b}\frac{\partial x^p}{\partial x'^c}\Gamma^m_{np} + \frac{\partial x'^a}{\partial x^m}\frac{\partial^2 x^m}{\partial x'^b \partial x'^c} \\
\implies g'^{bc}\Gamma'^a_{bc} &= g'^{bc}\frac{\partial x'^a}{\partial x^m}\frac{\partial x^n}{\partial x'^b}\frac{\partial x^p}{\partial x'^c}\Gamma^m_{np} + g'^{bc}\frac{\partial x'^a}{\partial x^m}\frac{\partial^2 x^m}{\partial x'^b \partial x'^c} \\
&\text{[contracting both sides by } g'^{bc}\text{]}
\end{aligned}$$

$$
\begin{aligned}
\implies \Gamma'^a &= \frac{\partial x'^a}{\partial x^m} g^{np} \Gamma^m_{np} + g'^{bc} \frac{\partial x'^a}{\partial x^m} \frac{\partial^2 x^m}{\partial x'^b \partial x'^c} \\
\implies \Gamma'^a &= \frac{\partial x'^a}{\partial x^m} \Gamma^m + g'^{bc} \frac{\partial x'^a}{\partial x^m} \frac{\partial^2 x^m}{\partial x'^b \partial x'^c} \\
\implies \Gamma'^a &= \frac{\partial x'^a}{\partial x^m} \Gamma^m + g'^{bc} \left(\frac{\partial}{\partial x'^b} \left(\frac{\partial x'^a}{\partial x^m} \frac{\partial x^m}{\partial x'^c} \right) - \frac{\partial^2 x'^a}{\partial x'^b \partial x^m} \frac{\partial x^m}{\partial x'^c} \right) \\
\implies \Gamma'^a &= \frac{\partial x'^a}{\partial x^m} \Gamma^m + g'^{bc} \left(\frac{\partial}{\partial x'^b} \left(\delta^a_c \right) - \frac{\partial^2 x'^a}{\partial x'^b \partial x^m} \frac{\partial x^m}{\partial x'^c} \right) \\
\implies \Gamma'^a &= \frac{\partial x'^a}{\partial x^m} \Gamma^m - g'^{bc} \frac{\partial^2 x'^a}{\partial x'^b \partial x^m} \frac{\partial x^m}{\partial x'^c} \\
\implies \Gamma'^a &= \frac{\partial x'^a}{\partial x^m} \Gamma^m - g'^{bc} \frac{\partial x^n}{\partial x'^b} \frac{\partial^2 x'^a}{\partial x^n \partial x^m} \frac{\partial x^m}{\partial x'^c} \\
\implies \Gamma'^a &= \frac{\partial x'^a}{\partial x^m} \Gamma^m - g^{mn} \frac{\partial^2 x'^a}{\partial x^m \partial x^n}
\end{aligned}
$$

So, the new coordinate system will satisfy the gauge condition (i.e. $\Gamma'^a = 0$) if we can find four functions $x'^a(x^m)$ that solve the four differential equations,

$$
\frac{\partial x'^a}{\partial x^m} \Gamma^m = g^{mn} \frac{\partial^2 x'^a}{\partial x^m \partial x^n}
$$

It is claimed without proof that solutions to these PDEs exist.

Now, Paolo Pani textbook claims that the name “harmonic" comes from the fact that under this gauge, $\nabla_a \nabla^a \Phi = 0$ becomes $\partial_a \partial^b \Phi = 0$. Why is that called harmonic I do not know.[Ask], but in the meantime let me show that it is a valid claim.

$$
\begin{aligned}
& \nabla_a \nabla^a \Phi = 0 \\
\implies & g^{ab} \nabla_a \nabla_b \Phi = 0 \\
\implies & g^{ab} \left(\partial_a \nabla_b \Phi - \Gamma^m_{ba} \nabla_m \Phi \right) = 0 \\
\implies & g^{ab} \partial_a \nabla_b \Phi - \Gamma^m \nabla_m \Phi = 0 \\
\implies & g^{ab} \partial_a \nabla_b \Phi - 0 = 0 \\
\implies & g^{ab} \partial_a \partial_b \Phi = 0 \quad \text{(For a scalar, } \nabla_b \Phi = \partial_b \Phi\text{)} \\
\implies & \partial_a \partial^a \Phi = 0
\end{aligned}
$$

5.2 Perturbative Approach

Let $\overset{0}{g}_{ab}$ is a known solution to the EFE. Let $\overset{0}{T}_{ab}$ be the stress-energy tensor sourcing this solution. Now, perturb $\overset{0}{T}_{ab}$ by δT_{ab} , i.e. $T_{ab} = \overset{0}{T}_{ab} + \delta T_{ab}$. This perturbation

changes the metric from $\overset{0}{g}_{ab}$ to $g_{ab} = \overset{0}{g}_{ab} + h_{ab}$ where $|h_{ab}| << \left|\overset{0}{g}_{ab}\right|$.

Now, I define the upper index h^{ab} as $h^{ab} = \overset{0}{g}^{ma}\overset{0}{g}^{nb}h_{mn}$. That is, h_{ab} is such that its indices (I do not know if calling h a tensor is justified. People seem to call it a tensor wrt the background metric. But if I am following the rule of tensor transformation, it is definitely not clear that h follows that rule, even when I am going from one coordinate system to another that follows the same metric structure of background+small perturbation. I will need to ask about this.) can be made up and down with the background metric. With this definition, the upper index g^{ab} is, up to $O(h)$, $g^{ab} = \overset{0}{g}^{ab} - h^{ab} + O(h^2)$ A quick proof of this is to see if $g^{ab}g_{bc}$ is equal to δ^a_c up to order h.

$$\begin{aligned} g^{ab}g_{bc} &= \left(\overset{0}{g}^{ab} - h^{ab}\right)\left(\overset{0}{g}_{bc} - h_{bc}\right) \\ &= \delta^a_c + h^a{}_c - h^a{}_c + O(h^2) \\ &= \delta^a_c + O(h^2) \end{aligned}$$

To be clear, EFE are invariant under arbitrary coordinate transformations. However, in this perturbative framework, only those transformations are allowed that allows us to write the metric as $g_{ab} = \overset{0}{g}_{ab} + h_{ab}$ where $|h_{ab}| << \left|\overset{0}{g}_{ab}\right|$. Now, I would like to find the equation that governs h_{ab}. The tentative plan is this, I will expand the left and right hand sides of EFE up to order h, and ignore the higher order terms. Then the background metric(i.e. Ricci tensor, $\overset{0}{R}_{ab}$) in left hand side will correspond to the background stress energy tensor, $\overset{0}{T}_{ab}$ and since we have assumed that the backgrond metric is the solution of the background EFE, I will simply remove those parts from LHS and RHS. The rest will be an equation for h_{ab}.

Now there are three necessary ingredients that I want to build and expand up to $O(h)$ as these are building blocks of EFE.

- $\Gamma^a{}_{bc}$ up to $O(h)$
- R_{ab} up to $O(h)$
- T up to $O(h)$

$\Gamma^a{}_{bc}$ up to $O(h)$

$$\Gamma^a_{bc}(g_{mn}) = \frac{1}{2}g^{am}\left(-\partial_m g_{bc} + \partial_b g_{cm} + \partial_c g_{mb}\right)$$

$$
\begin{aligned}
&= \frac{1}{2}\left(\overset{0}{g}^{am} - h^{am}\right)\left(-\partial_m \overset{0}{g}_{bc} + \partial_b \overset{0}{g}_{cm} + \partial_c \overset{0}{g}_{mb} - \partial_m h_{bc} + \partial_b h_{cm} + \partial_c h_{mb}\right) \\
&= \overset{0}{\Gamma}{}^{a}_{bc} + \frac{1}{2}\overset{0}{g}^{am}\left(-\partial_m h_{bc} + \partial_b h_{cm} + \partial_c h_{mb}\right) - \frac{1}{2}h^{am}\left(-\partial_m \overset{0}{g}_{bc} + \partial_b \overset{0}{g}_{cm} + \partial_c \overset{0}{g}_{mb}\right) + O(h^2) \\
&= \overset{0}{\Gamma}{}^{a}_{bc} + \delta\Gamma^{a}{}_{bc} + O(h^2)
\end{aligned}
$$

where $\delta\Gamma^{a}{}_{bc}$ is defined by ,

$$
\delta\Gamma^{a}{}_{bc} = \frac{1}{2}\overset{0}{g}^{am}\left(-\partial_m h_{bc} + \partial_b h_{cm} + \partial_c h_{mb}\right) - \frac{1}{2}h^{am}\left(-\partial_m \overset{0}{g}_{bc} + \partial_b \overset{0}{g}_{cm} + \partial_c \overset{0}{g}_{mb}\right)
$$

$\delta\Gamma^{a}{}_{bc}$ can be thought of as the variation of the Christoffel symbol Γ^{a}_{bc} when the metric $\overset{0}{g}_{mn}$ is varied by $\delta g_{mn} = h_{mn}$.In that sense, $\delta\Gamma^{a}{}_{bc}$ is a tensor even though Γ^{a}_{bc} is not. Rigorously understand this statement.

R_{ab} up to $O(h)$

In the convention that I am following, Riemann tensor is given by,

$$
R^{m}{}_{npq} = \partial_p \Gamma^{m}_{nq} - \partial_q \Gamma^{m}_{np} + \Gamma^{m}_{rp}\Gamma^{r}_{nq} - \Gamma^{m}_{rq}\Gamma^{r}_{np}
$$

Now, what is the Ricci tensor, R_{ab} look like?

$$
\begin{aligned}
R^{m}{}_{apb} =& \partial_p \Gamma^{m}_{ab} - \partial_b \Gamma^{m}_{ap} + \Gamma^{m}_{rp}\Gamma^{r}_{ab} - \Gamma^{m}_{rb}\Gamma^{r}_{ap} \\
R_{ab} =& R^{m}{}_{amb} = \partial_m \Gamma^{m}_{ab} - \partial_b \Gamma^{m}_{am} + \Gamma^{m}_{rm}\Gamma^{r}_{ab} - \Gamma^{m}_{rb}\Gamma^{r}_{am}
\end{aligned}
$$

Now, let us expand R_{ab} up to order h.

$$
\begin{aligned}
R_{ab} =& \partial_m \Gamma^{m}_{ab} - \partial_b \Gamma^{m}_{am} + \Gamma^{m}_{rm}\Gamma^{r}_{ab} - \Gamma^{m}_{rb}\Gamma^{r}_{am} \\
R_{ab} =& \partial_m\left(\overset{0}{\Gamma}{}^{m}_{ab} + \delta\Gamma^{m}{}_{ab} + O(h^2)\right) - \partial_b\left(\overset{0}{\Gamma}{}^{m}_{am} + \delta\Gamma^{m}{}_{am} + O(h^2)\right) \\
&+ \left(\overset{0}{\Gamma}{}^{m}_{rm} + \delta\Gamma^{m}{}_{rm} + O(h^2)\right)\left(\overset{0}{\Gamma}{}^{r}_{ab} + \delta\Gamma^{r}{}_{ab} + O(h^2)\right) \\
&- \left(\overset{0}{\Gamma}{}^{m}_{rb} + \delta\Gamma^{m}{}_{rb} + O(h^2)\right)\left(\overset{0}{\Gamma}{}^{r}_{am} + \delta\Gamma^{r}{}_{am} + O(h^2)\right)
\end{aligned}
$$

$$= \left(\partial_m \overset{0}{\Gamma}{}^m_{ab} - \partial_b \overset{0}{\Gamma}{}^m_{am} + \overset{0}{\Gamma}{}^m_{rm} \overset{0}{\Gamma}{}^r_{ab} - \overset{0}{\Gamma}{}^m_{rb} \overset{0}{\Gamma}{}^r_{am} \right)$$
$$+ \left(\partial_m \delta\Gamma^m{}_{ab} - \partial_b \delta\Gamma^m{}_{am} + \overset{0}{\Gamma}{}^m_{rm} \delta\Gamma^r{}_{ab} + \delta\Gamma^m{}_{rm} \overset{0}{\Gamma}{}^r_{ab} - \overset{0}{\Gamma}{}^m_{rb} \delta\Gamma^r{}_{am} - \delta\Gamma^m{}_{rb} \overset{0}{\Gamma}{}^r_{am} \right)$$
$$+ O(h^2)$$

In the last equality, the first term in parentheses is just $\overset{0}{R}_{ab}$. The rest is,

$$\text{second parentheses} = \partial_m \delta\Gamma^m{}_{ab} - \partial_b \delta\Gamma^m{}_{am} + \overset{0}{\Gamma}{}^m_{rm} \delta\Gamma^r{}_{ab} + \delta\Gamma^m{}_{rm} \overset{0}{\Gamma}{}^r_{ab} - \overset{0}{\Gamma}{}^m_{rb} \delta\Gamma^r{}_{am} - \delta\Gamma^m{}_{rb} \overset{0}{\Gamma}{}^r_{am}$$
$$= \left(\partial_m \delta\Gamma^m{}_{ab} + \overset{0}{\Gamma}{}^m_{rm} \delta\Gamma^r{}_{ab} - \delta\Gamma^m{}_{rb} \overset{0}{\Gamma}{}^r_{am} \right) - \left(\partial_b \delta\Gamma^m{}_{am} + \overset{0}{\Gamma}{}^m_{rb} \delta\Gamma^r{}_{am} - \delta\Gamma^m{}_{rm} \overset{0}{\Gamma}{}^r_{ab} \right)$$

The first and second parentheses look like background covariant derivatives of $\delta\Gamma^m{}_{ab}$ and $\delta\Gamma^m{}_{am}$, but we are missing one term in each parenthesis. so we add and substract that missing term.

$$= \left(\partial_m \delta\Gamma^m{}_{ab} + \overset{0}{\Gamma}{}^m_{rm} \delta\Gamma^r{}_{ab} - \overset{0}{\Gamma}{}^r_{am} \delta\Gamma^m{}_{rb} - \overset{0}{\Gamma}{}^r_{bm} \delta\Gamma^m{}_{ar} \right)$$
$$- \left(\partial_b \delta\Gamma^m{}_{am} + \overset{0}{\Gamma}{}^m_{rb} \delta\Gamma^r{}_{am} - \overset{0}{\Gamma}{}^r_{ab} \delta\Gamma^m{}_{rm} - \overset{0}{\Gamma}{}^r_{mb} \delta\Gamma^m{}_{ar} \right)$$
$$= \overset{0}{\nabla}_m \delta\Gamma^m{}_{ab} - \overset{0}{\nabla}_b \delta\Gamma^m{}_{am}$$

So up to O(h) R_{ab} is given by $R_{ab} = \overset{0}{R}_{ab} + \overset{0}{\nabla}_m \delta\Gamma^m{}_{ab} - \overset{0}{\nabla}_b \delta\Gamma^m{}_{am}$

T up to $O(h)$

In this context of perturbation theory, $T_{ab} = \overset{0}{T}_{ab} + \delta T_{ab}$. So,

$$T = g^{ab} T_{ab}$$
$$= \left(\overset{0}{g}{}^{ab} - h^{ab} \right) \left(\overset{0}{T}_{ab} + \delta T_{ab} \right)$$
$$=$$